

Research Report on

domain-gated workflows turning physics-based metamorphic relations into auditable V&V tests

Full search query: I want to find whether prior work in metamorphic testing, scientific software testing, machine-learning testing, and scientific verification & validation for PDE/CFD/physics-simulation surrogates has already established a domain-validity-gated workflow that converts physics-derived candidate metamorphic relations into auditable, executable V&V assets—specifically through measurement-floor admissibility conditions, explicit MR specifications/cards, and typed verdict records—and, if not, how to position this contribution distinctly from existing work on the test oracle problem, MR identification/generation, MR applicability constraints, MT for ML, and complementary SciML validation methods such as residual diagnostics, uncertainty quantification, conformal prediction, trust-region methods, and certified error bounds

Summary

No existing work defines a full domain validity gated workflow that turns physics derived metamorphic relations for PDE/CFD/physics simulation surrogates into auditable test assets with measurement floor admissibility, explicit MR “cards,” and typed verdict records; existing papers provide only partial ingredients, notably generic MR schemas and constraints [1,2], MR violation interpretation [3], simulation/scientific MT frameworks [8,9,10], and physics based MT for ML surrogates in hydrology and CFD like settings [4,5,6,7,9], but they never combine these elements into the integrated V&V pipeline you are envisioning.

Overall Answer to the Search Question

No paper among [1,2,3,4,5,6,7,8,9,10] implements all four of your target features together for physics based PDE/CFD/SciML surrogates:

1. Domain validity gating / regime constraints attached to each MR and used to gate evaluation.
1. Measurement floor or error budget admissibility conditions that can render tests inadmissible/inconclusive.
1. Explicit MR specifications (“MR cards”) as reusable, machine readable, auditable artifacts.
1. Typed verdict records (pass/fail/inconclusive/unsupported) with structured evidence.

Instead, the literature breaks into **complementary but disjoint strands**:

- **General MT infrastructure and MR schemas/constraints** [1,2,3].
- **MT for simulation and scientific/nuclear numerical codes** [8,9,10,9].
- **MT for ML/surrogate models in physical domains (hydrology, fluid dynamics)** [4,5,6,7].

These strands give you **raw materials**—physics derived MRs, generic MR representations, notions of MR applicability, and simulation/ML validation use cases—but **none of them assemble a domain validity gated, error budget aware, typed verdict workflow or integrate SciML diagnostics as MR gates**.

Key Pieces of Prior Work and How They Map to Your Target Workflow

1. MR Representation, Applicability Constraints, and Violation Interpretation

What they provide

- **Standardized MR representation and selection/constraint mechanisms**
 - **Duque Torres 2024** proposes a **comprehensive MR repository** and a **standardized, machine readable MR representation**, plus MetraTrimmer for test data driven **selection and constraining** of MRs [1].
 - **Duque Torres & Pfahl 2023** outline a “**complete MT pipeline**” with:
 - A DSL for describing MRs;
 - An explicit **MR constraint definition phase** to capture applicability;
 - A focus on explainability of MR outcomes [2].
- **Interpretation of MR violations (bug vs. bad MR/data)**
 - **Duque Torres et al. 2023** analyze **reasons behind MR violations**, using association rule mining to distinguish violations likely due to **bugs** from those due to **MR misfit** for specific test data [3].

What is missing relative to your workflow

- Constraints are **generic** (logical conditions, data space heuristics), not **physics regime constraints** (e.g., Reynolds/Mach ranges, laminar vs. turbulent) and are not tied to numerical error budgets [1,2].

- The MR schema does **not** include:
 - Measurement floor/error budget fields,
 - Links to PDE residuals, trust region indicators, or UQ.
- Violation analysis in [3] suggests a *conceptual* distinction between “bug” vs. “MR not applicable,” but it is **not formalized as typed verdict records** with explicit categories like *pass/fail/inconclusive/unsupported*.

Implication for positioning

- You can **explicitly build on** [1,2] as the closest prior art for MR “cards” and applicability constraints, and on [3] for the idea that outcomes need richer semantics, while pointing out that:
 - Prior work is domain agnostic and data driven,
 - It does not integrate physics based regime constraints, error budgets, or SciML diagnostics,
 - And it never defines a **typed verdict schema** as part of the MT artifact model.

2. MT for Simulation and Scientific/Nuclear Numerical Codes

Simulation validation and scientific codes

- **Chen et al. 2002:** *Metamorphic testing of programs on partial differential equations* [10].
 - Early demonstration that **PDE properties** can be encoded as MRs to test numerical codes without exact solutions.
- **Olsen & Raunak 2019:** *Increasing Validity of Simulation Models Through MT* [8].
 - Reframes MT as a method for **simulation validation**, using MRs as pseudo oracles for ABM and DES models when no oracle exists.
- **Li et al. 2022:** *Lightweight verification for nuclear power software based on MRs* [9].
 - Derives MRs systematically from **governing physical equations** and **numerical algorithms** (symmetry, homogeneity, error convergence behaviors), applied across unit/integration/system levels.

What they provide

- Strong evidence that **physics derived MRs**(symmetry, conservation, convergence, etc.) are viable and useful for:
 - PDE codes [10],
 - General executable simulations [8],
 - Safety critical numerical software [9].
- Clear **simulation V&V framing**—not just code correctness, but model behavior plausibility [8].

What is missing relative to your workflow

- MRs are typically specified **informally** (in prose or embedded in test code); there is:
 - No MR “card” with a standardized schema,
 - No explicit per MR **regime of validity encoding**(e.g., nondimensional parameter bounds or discretization regimes).
- No explicit modeling of **measurement floors or discretization error budgets** as admissibility conditions for MR evaluation.
- Verdicts are essentially **binary** (MR satisfied/violated), without typed verdict semantics or explicit “inconclusive/unsupported” states.

Implication for positioning

- These works show that **your idea is not about inventing physics based MRs** but about:
 - Formalizing them into auditable MR assets,
 - Enriching them with **regime constraints and error budgets**, and
 - Introducing **typed verdicts**—all absent from current simulation MT/V&V practice.

3. MT for ML / Surrogate Models in Physical Domains (SciML adjacent)

Hydrologic ML models

- **Yang & Chui 2021:** *Reliability assessment of ML models in hydrological predictions through MT* [7].
 - Defines hydrologic MRs (e.g., **runoff monotonicity with rainfall**), applied to ML flood predictors where ground truth for future events is unavailable.
 - Shows **prediction accuracy and MR consistency can be uncorrelated**, emphasizing MT’s added value.

- **Reichert et al. 2024:** *Metamorphic testing of machine learning and conceptual hydrologic models* [6].
 - Extends MT to **multiple models and calibrations**; uses MRs plus **sensitivity analysis** to training/calibration to study model behavior under changing drivers.

CFD like ML surrogates and physical field prediction

- **Qi et al. 2025:** *Multi scenario collaborative testing methods for intelligent algorithms in physical field prediction*[4].
 - Multi scenario framework (benchmark, extreme, perturbation, cross domain) for ML velocity field prediction around a cylinder.
 - Uses **constraint rule coupled MT**, constructing MRs from **multi parameter physical correlations** and experimental statistical patterns; evaluates functional correctness, physical consistency, extrapolation capability, and real time performance [4].
- **Yu et al. 2025:** *Mutation testing strategies for intelligent models predicting fluid velocity fields* [5].
 - Designs **disturbance and cross domain scenarios**, MRs based on **Reynolds number conservation** and **Strouhal number correlations**.
 - Uses MT and mutation testing to quantitatively evaluate **physical consistency** and **extrapolation capability** of ML velocity field surrogates [5].

What they provide

- Clear **precedent for physics derived MRs applied to ML/surrogate models**
 - Hydrologic monotonicity/physical expectations [7,6].
 - Dimensionless number based relations (Re, Strouhal) and physical correlations for CFD like ML models [4,5].
- Emphasis on **extrapolation and physical consistency**:
 - Scenario taxonomies (benchmark/extrapolation) [4,5].
 - Analysis of **model reliability beyond observed data** [7,6].
- Strong alignment with your application domain (hydrology, CFD like flows, physical fields).

What is missing relative to your workflow

- MRs remain **informal or lightly specified**; there is no explicit, standardized **MR card** schema, and regime information is tied to scenarios or problem statements rather than encoded per MR.
- Scenarios (e.g., “cross domain”, “extreme”) act as **high level operational envelopes** but they are not formalized into per MR **domain validity gates**
- No explicit **measurement floor admissibility**
 - While experimental/statistical patterns are used [4], they are not translated into error budgets that determine whether MR deviations are interpretable.
 - Sensitivity to training/calibration [6] is analyzed, but not used to gate MR tests.
- Verdicts are either:
 - Qualitative (consistent/inconsistent) or metric based (quality scores, mutation scores), or
 - At most interpretive distinctions, but **not typed verdict records** with fields for “out of domain” or “inconclusive due to high uncertainty.”

Implication for positioning

- These are your **closest domain specific neighbors** MT for ML surrogates in hydrology and CF like settings, using physics based MRs for extrapolation and physical consistency.
- You can argue that **your novelty is not in the MR content** (Reynolds/Strouhal, monotonicity, etc.), but in:
 - Embedding such MRs in an explicit schema with regime constraints and measurement floor gates;
 - Using SciML diagnostics (residuals, UQ, trust region flags) to decide when these MRs are **admissible** and how to **type** verdicts.

4. Integration with SciML Validation Tools (Residuals, UQ, Trust Regions, Certified Bounds)

Across all surveyed papers [1,2,3,4,5,6,7,8,9,10], none:

- Use **PDE residual norms**, physics informed loss terms, or PDE violation diagnostics to **gate MR tests** or to type verdicts.
- Integrate **uncertainty quantification**, conformal prediction, or predictive intervals into MR predicates (e.g., comparing or enforcing consistency of intervals instead of point outputs).
- Use **trust region or extrapolation detectors** as explicit signals that render certain MR evaluations “unsupported/out of domain”.
- Tie **certified error bounds** (e.g., a posteriori error estimates for ROMs/PDE surrogates) to MR tolerance selection or admissibility conditions.

Where uncertainty/sensitivity appears:

- In [6], **multi model and calibration sensitivity** is used to interpret differences but not as formal admissibility logic.
- In [4,5,7], the focus is on **scenario based evaluation** and error metrics, not on integrating diagnostics as gates.

Implication for positioning

- You can legitimately claim that using **SciML diagnostics as first class gating and typing signals for MR based V&Vs not present** in current literature.
- This differentiates your workflow from:
 - Pure MT for ML approaches [4,5,6,7], and
 - Classical simulation V&V with MT [8,9,10].

How to Position a New Contribution

Based on the above, you can position your proposed **domain validity gated workflow for physics derived MRs in SciML/PDE/CFD surrogates** as follows:

1. Build Explicitly on Three Existing Pillars

1. Physics derived MRs for scientific codes and surrogates

- Acknowledge that using **physical invariants, symmetries, conservation laws, and dimensionless relations** as MRs is well established:
 - PDE/numerical codes [10,9],
 - Simulation models [8],
 - ML surrogates in hydrology and CFD like physical fields [7,6,4,5].

1. Formal MR representation and constraints

- Cite [1,2] for the idea of **machine readable MRs** MR repositories, DSLs, and **applicability constraints**, and [3] for interpreting MR violations.

1. MT as a tool for validation and reliability assessment

- Use [8] to anchor your work in **simulation validation**, and [7,6,4,5] to ground the application to **ML surrogates where ground truth is partial or absent**.

2. Highlight the Novel Integrative Aspects

Emphasize that **no prior work** jointly provides:

- **MR cards for physics/CFD/SciML** that:
 - Encode input/transform signatures and expected predicates,
 - Attach **regime of validity constraints** derived from physics and numerical analysis (e.g., non dimensional parameter ranges, mesh/time step bounds, closure/training regimes),
 - Include **measurement floor/error budget fields** (discretization error, solver tolerances, experimental noise, surrogate UQ),
 - Record traceability to governing equations, derivations, and model configurations.
- **Measurement floor admissibility logic** that:
 - Treats MR evaluation as **conditionally testable**, depending on whether the error floor makes the expected effect of the metamorphic transformation distinguishable,
 - Allows MR tests to be classified as **inconclusive** when the floor is too high, not simply “failed” or “passed.”
- **Typed verdict records** that:
 - Distinguish **pass, fail (with satisfied preconditions), inconclusive (due to uncertainty/measurement floor), and unsupported (out of domain or MR inapplicable)**
 - Store evidence: MR residuals, regime checks, and relevant SciML diagnostics,
 - Enable post hoc scientific auditability—separating genuine physical/numerical defects from out of regime or noise dominated cases.
- **Integration with SciML diagnostics** in the MT loop:
 - Using **residuals, UQ, trust region indicators, and certified error bounds**
 - To gate whether an MR is evaluated,
 - To shape admissible tolerances,

- And to influence verdict typing.

3. Clarify Distinctions from Related Themes

When writing, explicitly differentiate your contribution from:

- **Oracle construction / MR discovery work** that stops at identifying or generating MRs without turning them into structured, reusable test assets [1,2,3,10,9].
- **MT for ML** that uses physics based MRs but:
 - Does not formalize domain validity or error budgets,
 - Treats outcomes as binary (consistent/inconsistent) [7,6,4,5].
- **SciML validation methods (residuals, UQ, conformal prediction, trust regions)** used in isolation for model quality or calibration, not as part of a **unified MT based V&V workflow**

Bottom Line for Your Research Direction

The literature confirms that:

- **The ingredients you want to use are well motivated and independently established**
 - Physics derived, regime dependent MRs [10,8,9,7,6,4,5].
 - Generic MR schemas and applicability constraints [1,2].
 - Interpretation of MR violations beyond simple pass/fail [3].
 - SciML diagnostics for error/uncertainty and extrapolation (implicitly in [6,7,4,5]).

But **no paper** combines these into a **domain validity gated, measurement floor aware MT workflow** with **explicit MR cards and typed verdict records** for PDE/CFD/physics simulation surrogates. That integration—especially the explicit admissibility logic and tight coupling to SciML diagnostics—is the clear conceptual and methodological space where your contribution will be distinct.

Categories

Comparative Overview of How the Papers Address Domain Valid, Auditable MT Workflows

Compared dimensions and rationale

To make the comparison meaningful for your specific topic, the table and discussion below focus on whether and how each paper addresses the **four core workflow elements** you care about, plus closely related aspects:

1. **Physics derived metamorphic relations (MRs)**– Are MRs grounded in physical laws or models (PDE/CFD/physics/empirical hydrology, etc.)?
1. **Domain validity gating / regime constraints**– Are there *explicit, operational* conditions under which an MR is applicable (e.g., parameter ranges, regimes), and are they actually used to *gate* MR evaluation?
1. **Measurement floor / error budget admissibility**– Are numerical/measurement uncertainties explicitly modeled as a *precondition* for interpreting MR outcomes (i.e., tests can be “inadmissible” if error floor too large)?
1. **Explicit MR specifications / “MR cards”**– Is there anything like a **structured MR schema** (inputs, transformation, expected predicate, validity constraints, metrics, traceability), beyond informal prose or code?
1. **Typed verdict records (beyond pass/fail)**– Does the paper distinguish **different types of outcomes** for MR checks (e.g., pass / fail / inconclusive / out of domain) in a structured way?
1. **SciML / surrogate models for PDE/physics**– Does the work target ML/surrogate models for physical systems (e.g., hydrologic ML, physical field prediction) rather than only conventional software or simulations?
1. **Integration with other SciML validation tools**– Are residual diagnostics, UQ, conformal prediction, trust regions, or certified bounds integrated *with MR evaluation* (e.g., as gating conditions or in verdict typing)?
1. **Oracle problem positioning / V&V framing**– Does the paper explicitly frame its approach in terms of the test oracle problem or scientific V&V (simulation validation vs. code verification vs. ML reliability)?

These dimensions are summarized in the table, then elaborated thematically.

Tabular Comparison of Key Aspects

Table 1 – Alignment of Each Paper with the Desired Domain Valid, Auditable MT Workflow

Legend (per column): Y = clearly present; P = partial / implicit / limited; N = not present in the available text.

Ref	Domain / target	Physics derived MRs	Domain validity gating / regime constraints	Measurement floor / error budget admissibility	Explicit MR schema / "cards"	Typed verdicts (non binary)	MT for ML / physical surrogates	Integration with residuals / UQ / trust regions	Oracle / V&V framing
[1]	General MT across domains	P – repository of MRs, not physics specific	P – focuses on constraining applicability, but not PDE/physics regimes	N – no explicit error budget admissibility in excerpts	Y – standardized, machine readable MR representation	N – violations treated as binary outcomes in description	N	N	Y – framed in test oracle / MR selection context
[2]	General MT pipeline	P – MRs from various domains, not specifically physics	Y (conceptual) – MR constraint definition phase for applicability [2]	N	P – domain specific language to describe MRs, not yet a full "card"	P – discusses "explainability of MR outcomes" but not typed verdict schema	N	N	Y – positioned as MT pipeline, oracle problem context
[3]	General MT (toy examples)	N – focus on generic software MRs	P – analyzes when MR "does not fit test - data conditions" [3]	N	P – refined MRs, but no explicit schema	P – conceptually distinguishes bug vs. MR misfit, but not as structured verdict types	N	N	Y – explicitly about interpreting MR violations, oracle issues
[4]	Intelligent physical - field prediction (e.g. flow around cylinder)	Y – MRs via multi parameter physical correlations and experimental patterns [4]	P – scenario taxonomy (benchmark / extreme / perturbation / cross domain) implies regimes, but not formal gating per MR	N (from excerpt) – no explicit measurement floor/error budgets	N – MRs described conceptually, no MR schema mentioned	P – "multi attribute evaluation" scores, but not typed MT verdicts per MR	Y – ML for physical field prediction	N (from excerpt) – relies on scenarios and MT, not residual/UQ integration	P – V&V like multi scenario assessment, but not detailed oracle discussion
[5]	Intelligent models for fluid velocity fields	Y – MRs based on Reynolds and Strouhal number relations [5]	P – disturbance vs cross domain scenarios hint at operational envelopes, not formal MR gating	N	N	N – evaluation is quantitative but described as generic	Y – ML for fluid velocity prediction	N – focuses on MT and mutation testing, no residual/UQ gating in excerpt	P – talks about limitations of traditional testing and physical consistency, but not explicit oracle framing
[6]	Hydrologic ML + conceptual models	Y – MRs derived from hydrologic expectations (e.g., monotonicity) [6]	P – some expectations hold only for certain perturbations but regimes not formalized as gates	N – no explicit measurement floor admissibility in the excerpt	N – MRs mostly in textual form	P – distinctions like "consistent" vs. "inconsistent" predictions in MT analysis, but not a typed schema	Y – ML + conceptual hydrologic models	P – includes sensitivity analysis to training/calibration options, but not used to gate MR tests	Y – explicitly about assessing reliability where no oracle exists
[7]		Y – MRs like "runoff in-	P – some discussion of similarity to observed data	N	N	P – conceptually distinguishes different		N – does not use UQ/trust regions as gates in	Y – explicitly frames MT as deal-

	Hydrologic ML (flood magnitude)	creases with rainfall" [7]	affecting assessment; not an explicit validity gate			factors for inconsistent predictions, but verdicts are essentially pass/fail on MR consistency	Y – MT for ML hydrologic models	provided text	ing with absent oracles
[8]	Simulation models (ABM, DES)	Y – MRs from parameters/behaviors of executable models [8]	P – guidelines for applying MRs for validation, but no regime of validity formalization	N – no explicit measurement floor or error budgets in the excerpt	N – guidelines, not a machine-readable MR card	N – examples suggest binary MR satisfaction	N – not ML surrogates, but simulations	N – does not integrate residual/UQ diagnostics as gates	Y – explicit simulation validation framing and oracle problem
[9]	Nuclear power numerical software	Y – MRs from governing physical equations and numerical algorithms [9]	P – hierarchical MR model hints at structured coverage, but regime constraints not explicit	N	N – hierarchical MR "model" but not a structured, reusable MR schema as in [1]	N – focuses on detecting coding/numerical errors; verdicts are effectively pass/fail	N – targets numerical code, not ML surrogates	N	Y – framed as light-weight verification and mitigation of oracle problem
[10]	PDE programs (classical numerical codes)	Y – MRs for PDE programs [10]	N/P – classic case study; likely mentions where MRs hold, but no explicit regime gating in the excerpt	N	N	N	N – PDE solvers, not ML surrogates	N	Y – foundational MT for PDE software, oracle framing implied

Key high level conclusion from the tableNone of the papers [1,2,3,4,5,6,7,8,9,10] implement all four of your core workflow elements together (domain validity gating, measurement floor admissibility, explicit MR cards, and typed verdict records) for physics derived MRs in SciML/PDE/CFD contexts. Several contribute **pieces** of the picture, but the integrated, domain validity gated workflow with auditable MR artifacts and typed verdicts does not appear as such.

Thematic Comparison Across the Papers

1. Physics derived MRs and scientific / PDE / CFD contexts

- **Strong physics based MR grounding– PDE/physics codes and simulations:**• [10] is the classic case study of MT for programs solving PDEs, deriving MRs from known properties of numerical PDE solutions. • [9] derives MRs from *governing physical equations* and numerical schemes for nuclear power software, including classes like symmetry, homogeneity, and error convergence [9]. • [8] derives MRs from parameters/behaviors of simulation models (ABM, DES), treating them as pseudo oracles for simulation *validation* [8].– **Physical surrogates / ML for physics:**• [4] constructs MRs based on “multi parameter physical correlations and experimental statistical patterns of key parameters” for intelligent physical field prediction [4]. • [5] explicitly uses Reynolds and Strouhal number relations as metamorphic constraints for ML models predicting fluid velocity fields [5]. • [6,7] define hydrologic MRs (e.g. monotonicity of runoff with rainfall) grounded in hydrological physics/empiricism [6,7].
- **More generic, non physics MR work**– [1,2,3] operate across domains: [1] maintains a large MR repository and a standardized representation; [2] designs a general MT pipeline; [3] analyzes MR violations via association rule mining.

Relative to your goal:The physics derived MR *content* exists and is reasonably mature for simulations [8,9,10] and for some ML surrogates in hydrology and fluid dynamics [4,5,6,7]. However, these works largely do **not** structure these MRs into auditable assets with explicit regime gating and admissibility conditions.

2. Domain validity gating and regime constraints

- **Closest to explicit applicability constraints (but still generic):**– [2] introduces an MR “constraint definition phase” to capture “conditions under which MRs hold,” as a key step in its envisioned MT pipeline [2]. This is the clearest *conceptual* acknowledgement that MRs must be constrained and that applicability needs to be represented explicitly, though not specifically for physics/CFD regimes, and without linking to numerical error budgets.
 - [1] also emphasizes that MRs are **not universally applicable** and develops techniques (MetraTrimmer) to “select and

constrain” MRs based on test data [1]. Constraints here are driven by data and ML/rule mining, but again not tied to physics based nondimensional parameters or SciML diagnostics.

- **Physics/scenario based notion of regimes, but not formal gates per MR**— [4] defines four scenario classes—benchmark, extreme, perturbation, cross domain—and uses these to organize validation and “physical extrapolation capability” tests for intelligent physical field predictors [4]. This is highly relevant at the *scenario level*, but the regime constraints are attached to scenarios rather than being encoded as part of each MR’s formal specification.
 - [5] defines “disturbance scenarios” and “cross domain scenarios” around Reynolds/Strouhal based relations [5]. These implicitly mark in domain vs. extrapolation conditions, but again, not as *formal MR preconditions* used to gate evaluation.
 - [6,7] sometimes interpret MR violations in light of input similarity to training data [7] or across different model structures and calibrations [6]. This approximates an implicit validity notion (“this MR check is more meaningful in certain regions of the input space”) but is not formalized as a gating mechanism.
- **Simulation oriented works [8,9,10]** acknowledge that MRs may have specific applicability, but the excerpts do not show that this is captured as a reusable, machine readable constraint attached to each MR (e.g., via nondimensional parameters, solver settings, or model regimes). The PDE case study [10] predates modern formal MT tooling, so this is unsurprising.

Relative to your workflow: You would need to go beyond [2]’s general constraint definition phase and [4,5,6,7]’s scenario/regime thinking by:

- Expressing regime constraints *per MR* (e.g., in MR cards), using physically meaningful parameters.
- Using those constraints as **hard gates** in MR evaluation, producing typed outcomes (e.g., “unsupported” when regime preconditions fail).

No paper in [1,2,3,4,5,6,7,8,9,10] does this in a fully explicit, domain validity gated way.

3. Measurement floor / error budget admissibility

- Across all papers [1,2,3,4,5,6,7,8,9,10], there is **no explicit notion** of:
 - Quantified numerical/measurement error budgets attached to MRs, **and**
 - A formal rule that an MR test is only *admissible* if the expected noise floor is below a discriminative threshold.
- Some works do implicitly acknowledge uncertainties or variability:
 - [6] includes a **sensitivity analysis** of predictions to training/calibration options as part of its MT-based assessment [6]. This recognizes that model variability matters when interpreting MR results, but does not cast this as a measurement floor gate for MR admissibility.
 - [7] explores how input similarity to observed data affects MT reliability assessments, hinting at a connection between epistemic uncertainty and MR interpretation [7], but without defining an explicit error budget or admissibility condition.
 - Simulation and numerical software papers [8,9,10] naturally operate in contexts where discretization and stochastic variability exist, yet the excerpts show no explicit allocation of error budgets to MR predicates.

Relative to your workflow: The *measurement floor admissibility* concept appears to be wholly absent. You would be adding a genuinely new dimension by:

- Explicitly estimating or bounding relevant numerical/experimental errors per run.
- Using these bounds to decide whether the MR residual can be meaningfully interpreted (pass/fail) or must be treated as **inconclusive**.

4. Explicit MR specifications / “MR cards”

- **Closest to an MR schema:**
 - [1] explicitly proposes a **standardized, machine readable MR representation** and a comprehensive MR repository [1]. This is the closest thing to your notion of an “MR card,” at least for generic software:
 - It supports collection and description of MRs in a standard schema.
 - It is not tied to physics/PDE surrogates, nor does it embed error budgets or SciML diagnostics.
 - [2] mentions a domain specific language (DSL) to “generate and describe MRs,” and a pipeline encompassing MR generation, applicability constraints, and explainability [2]. This suggests a structured specification, but details in the excerpt are limited.
- **Partial / implicit structures:**
 - [3] refines MRs through rule mining and distinguishes when MRs fail due to bugs vs. MR misfit [3]. This requires some structured notion of MR, but the excerpt does not indicate a full schema with preconditions, metrics, and traceability fields.
 - [9] describes a “hierarchical MR model” for nuclear power software [9], which organizes MRs but is not described as a machine readable schema akin to MR cards.
 - [4,5,6,7,8,9,10] typically specify MRs in natural language or embedded in test scripts/experimental design, not as reusable, inspectable artifacts with a standardized format.

Relative to your workflow: Building explicit, physics oriented MR cards for SciML/PDE surrogates would:

- Extend [1,2]'s generic MR schemas to include:
 - Regime-of-validity fields anchored in physical parameters.
 - Measurement-floor/error-budget fields.
 - Links to residual/UQ/trust-region diagnostics.
 - Bring the informal physics-based MRs of [4,5,6,7,8,9,10] into a **formalizable, auditable** representation.
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5. Typed verdict records and interpretation of MR outcomes

- **Structured interpretation of MR violations:**
 - [3] is the most directly relevant: it analyzes reasons behind MR violations and distinguishes whether a violation is more likely due to a **bug** or to the MR not fitting the test data [3]. This effectively introduces a conceptual distinction between:
 - *True failure* (bug),
 - *Invalid MR or out of scope data*
 However, this is not formalized as a **typed verdict schema** with fields for pass/fail/inconclusive/unsupported.
- **Non binary outcome semantics**(implicitly):
 - [2] mentions providing “explainability for MR outcomes” in its pipeline [2]. This suggests richer reporting, but the excerpt doesn’t show a verdict type system.
 - [4] uses a multi attribute scenario evaluation to assess performance across different scenarios and quality attributes [4]. This yields more nuanced performance characterizations, but not typed MR verdicts.
 - [6,7] often interpret MR consistency vs. inconsistency in the context of model reliability, distinguishing ML models that fit data well but violate MRs (suggesting unreliability under perturbations) [6,7]. Again, this is conceptually multi faceted evaluation, but the test outcomes themselves appear to be treated as binary (consistent/inconsistent).
- **Binary pass/fail in classical MT and V&V:**
 - [8,9,10,9] primarily use MR violations as classical failure indicators for simulations and numerical codes [8,9,10,9], without typed outcome notions like “inconclusive” or “unsupported.”

Relative to your workflow: You can clearly position the introduction of **typed verdict records** (pass / fail / inconclusive / unsupported, with evidence fields) as a novel contribution that:

- Generalizes [3]'s insight that MR violations may not always mean “bug.”
 - Makes this distinction *first class* and machine interpretable, especially for SciML surrogates where uncertainty and domain validity are crucial.
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6. MT for ML / SciML surrogates and interaction with other validation tools

- **MT for ML in physical domains:**
 - [7]: MT for ML models predicting flood magnitudes in hydrology; explicitly addresses the lack of ground truth for future events and uses MRs (e.g. monotonicity) to assess prediction reliability [7].
 - [6]: Extends MT to both ML and conceptual hydrologic models, adds multi model comparisons and sensitivity analysis to training/calibration [6].
 - [4]: Applies metamorphic testing to “intelligent physical field prediction algorithms” with scenario driven validation for physical consistency and extrapolation [4].
 - [5]: Uses metamorphic tests with physical constraints for intelligent models predicting fluid velocity fields, coupled with mutation testing to evaluate test effectiveness [5].
- **MT for simulations (non ML)**
 - [8] uses MT for executable simulation *validation* (ABM and DES), with MRs interpreted as pseudo oracles [8].
 - [9] and [10] address numerical codes for physics (nuclear power software; PDE programs) [9,10].
- **Integration with residual diagnostics, UQ, trust regions, etc.:**
 - In the available excerpts, **none** of the papers [1,2,3,4,5,6,7,8,9,10] explicitly:
 - Use residual norms or PDE consistency scores to **gate** whether MRs are evaluated or how verdicts are typed.
 - Integrate UQ or conformal prediction intervals into the MR predicates (e.g., checking interval consistency).
 - Use trust-region/extrapolation detectors to declare certain MR tests as out of domain/unsupported.
 - Tie certified error bounds to MR tolerances.

- The closest indirect themes are:
 - [6]'s use of multi model and calibration sensitivity analyses as auxiliary information when interpreting MT results [6].
 - [4]'s scenario driven multi attribute evaluation, which implicitly deals with extrapolation and robustness, though not via classical residual or UQ tools [4].

Relative to your workflow: The idea of a **domain validity gated MT workflow** that leverages SciML diagnostics (residuals, UQ, trust regions, error bounds) as *preconditions* for MR evaluation and as inputs to verdict typing appears to be missing entirely. This gives you a strong differentiation angle:

- Existing MT for ML works [4,5,6,7] treat MRs and other diagnostics largely **in parallel**, not as an integrated gating/verdict mechanism.
- Simulation validation works [8,9,10] do not exploit modern SciML style diagnostics.

Positioning Your Intended Contribution Relative to These Works

Pulling the comparison together:

- **MR content vs. MR workflow** Physics derived MRs are present and richly used in simulation [8,9,10,9] and in several ML for physics contexts [4,5,6,7]. However, these MRs are largely encoded informally or directly in tests, without:
 - Rigorous regime of validity encoding per MR,
 - Measurement floor admissibility logic,
 - Typed verdict records.
- **Closest analogues to MR “cards” and constraints**— [1] and [2] provide the strongest generic MT infrastructure: standardized MR representation and an MT pipeline with applicability constraints and explainable outcomes [1,2]. But they:
 - Are not tailored to physics/PDE/CFD surrogates,
 - Do not integrate measurement floors or SciML diagnostics,
 - Do not define typed verdict schemas.
- **Closest analogues to typed verdict distinctions**— [3] conceptually distinguishes MR violations caused by bugs vs. MR misfit [3], suggesting multi typed interpretations but not encoding them as reusable verdict types or linking them to domain validity and measurement floors.
- **Absence of measurement floor admissibility and SciML diagnostic gating**— Across [1,2,3,4,5,6,7,8,9,10], there is no explicit methodology that:
 - Computes or estimates error/uncertainty budgets,
 - Uses them as **preconditions** for MR evaluation,
 - Integrates residuals / UQ / trust regions into the logic of MR application and verdict typing.

Implication: Your proposed “domain validity gated workflow that converts physics derived candidate MRs into auditable, executable V&V assets—via measurement floor admissibility, MR cards, and typed verdict records, integrated with SciML diagnostics” appears to be **distinct** from existing work in the following ways:

1. **Level of formalization and auditability:** Extends [1,2]'s MR representation ideas to scientific/ML surrogates with rich, physics driven MR cards, including regime constraints and traceability to physical derivations.
1. **Explicit admissibility logic:** Introduces measurement floor/error budget conditions as first class preconditions for MR tests, something not present in [4,5,6,7,8,9,10].
1. **Typed verdict semantics:** Formalizes outcome types (pass / fail / inconclusive / unsupported) as data structures, generalizing the interpretive distinctions hinted at in [3] and implicitly in [4,6,7].
1. **Integration with SciML validation tools:** Uses residual diagnostics, UQ, trust regions, and certified error bounds not just for model selection or side analyses (as in some SciML work), but as **gating and typing signals** for MR based V&V assets—an integration absent in [1,2,3,4,5,6,7,8,9,10].

These comparisons should help you cite prior work accurately while clearly delineating the novel aspects of your proposed workflow.

Timeline

Chronological Evolution of Ideas Relevant to Domain Validity Gated MT for Physics/Surrogates

Early MT for PDE/Scientific Codes (2000s)

- 2002 – First explicit MT case study on PDE codes
-

Chen et al.'s case study on *metamorphic testing of programs on partial differential equations* [10] appears to be the earliest direct connection between MT and PDE based scientific software.

- **Conceptual contribution:**
 - Demonstrates that physics/numerics-based relations (e.g., mesh refinement effects, symmetry, parameter changes) can be encoded as MRs to test PDE solvers when exact solutions are unavailable.
- **Limitations relative to your workflow:**
 - MRs are used as *informal* partial oracles; there is no notion of:
 - Explicit regime of validity constraints attached to each MR,
 - Measurement floor/error budget admissibility,
 - MR “cards” as structured artifacts, or
 - Typed verdicts (pass vs. fail only, with no systematic “inconclusive/unsupported” categories).
- **Impact:**
 - Sets the precedent that MT is viable for numerical PDE programs, but at the level of manual MR design and classical fault detection.

MT for Simulation Validation (2010s)

- **2019 – General MT framework for executable simulation models**
 - Olsen & Raunak propose “Increasing Validity of Simulation Models Through Metamorphic Testing” [8].
- **Key step forward:**
 - Reframes MT from *software verification* to *simulation validation* for ABM and DES models.
 - Provides a **framework and guidelines** for deriving MRs from model structure and using them as pseudo oracles when no ground truth exists [8].
- **Relevance to your goal:**
 - Moves closer to the scientific-V&V viewpoint: MRs are not merely about program errors, but about whether the *simulation model* behaves plausibly as a representation of reality.
 - However, the framework:
 - Treats MRs as **test heuristics** rather than auditable MR assets with formal metadata.
 - Does not introduce domain validity gating (e.g., regime constraints tied to nondimensional parameters).
 - Uses tolerances heuristically rather than as derived error budgets; no explicit measurement floor admissibility.
 - Verdicts remain effectively binary (MR satisfied/violated), without explicit “inconclusive/unsupported” states.
- **Milestone:**
 - First widely cited work to position MT as a *validation* tool for scientific simulations, not just code verification.

Domain Specific MT for Scientific/Nuclear Numerical Codes (early 2020s)

- **2022 – MR based verification for nuclear power numerical software**
 - Li et al. propose a “lightweight verification method based on metamorphic relation for nuclear power software” [9].
- **Contributions:**
 - MRs are systematically derived from:
 - Governing physical equations,
 - Numerical algorithms, and
 - Program specifications [9].
 - Introduces a *hierarchical MR model* and MR categories such as symmetry, homogeneity, conjugation, and error - convergence behaviors.
- **Relative to your workflow:**
 - Strong step toward **physics derived MRs** in a safety critical domain.
 - Yet:
 - Regime of validity is implicit; constraints are not formalized as machine readable MR preconditions.
 - Numerical error and measurement uncertainty are not turned into explicit error budgets gating MR evaluation.
 - Verdicts are treated as “pass/fail” outcomes without typed classifications or explicit handling of inconclusive cases.

- **Trend:**
 - Shows growing interest in using MT as a lightweight alternative to high cost classical V&V for complex scientific codes, but still without the full domain validity gated workflow.

MT for ML Models in Physical / Environmental Domains (2020s)

- **2021 – MT for reliability assessment of ML hydrological models**
 - Yang & Chui apply MT to assess reliability of ML flood prediction models [7].
 - **Key ideas:**
 - MRs based on hydrological knowledge (e.g., monotonicity of runoff with respect to rainfall magnitude) are used to test predictions where ground truth is unavailable [7].
 - Shows that **prediction accuracy and MR based consistency are uncorrelated** highlighting MT's unique diagnostic value.
 - **Relation to your workflow:**
 - Strong example of **physics/phenomenology derived MRs for ML surrogates**
 - However:
 - MRs are specified qualitatively; no explicit MR schema with preconditions, admissible domains, or metrics.
 - No explicit measurement floor or uncertainty based gating (e.g., UQ or residuals driving test admissibility).
 - Verdicts are effectively binary: predictions are “consistent” or not, with no formal typing.
- **2024 – Multi model MT for ML and conceptual hydrologic models**
 - Reichert et al. extend the MT approach to multiple model structures and calibration options [6].
 - **Advances:**
 - MT is used in a **multi model comparative framework** checking consistency of qualitative expectations across different hydrologic model types and calibration strategies [6].
 - Incorporates **sensitivity analyses** of predictions to training/calibration options.
 - **Relevance and gap:**
 - Moves closer to **structured analysis** of MT outcomes (e.g., patterns across models), but:
 - Still no formal MR “card” representation with domain/regime constraints.
 - No explicit measurement floor admissibility—uncertainty and calibration variability are analyzed, not turned into MR gating conditions.
 - Verdicts are not explicitly typed (no systematic “inconclusive/unsupported” categorization).
- **Trend across [7,6]:**
 - MT is increasingly used as a **reliability and extrapolation assessment tool** for ML models in physical domains.
 - MRs are mostly **qualitative or semi quantitative** and the focus is on **model comparison and interpretive diagnostics**, not on formalizing MRs as reusable, domain gated V&V assets.

MT and Physical Constraints for ML Surrogates in CFD like Settings (mid 2020s)

- **2025 – Multi scenario MT for intelligent physical field prediction**
 - Qi et al. propose a **multi scenario collaborative testing framework** for velocity field prediction around a cylinder using intelligent algorithms [4].
 - **Notable features:**
 - Hierarchical scenario categories: **benchmark, extreme, perturbation, cross domain** scenarios target different attributes (functional correctness, stability, physical consistency, extrapolation, real time performance) [4].
 - **Constraint rule coupled MT:** MRs are built from multi parameter physical correlations and experimental statistical patterns, to test physical similarity and extrapolation [4].
 - A **scenario driven multi attribute evaluation** integrates quality attributes with scenario adaptation strategies.
 - **Relation to your target workflow:**
 - Closer to **operational envelope testing** of ML surrogates in a CFD like context.
 - Yet:
 - Regime constraints are expressed at the **scenario level**, not explicitly attached to each MR as machine checkable preconditions.
 - No explicit measurement floor or error budget gating—experimental/statistical patterns are used for evaluation, but not phrased as admissibility conditions that can render an MR “inconclusive”.

- Evaluation is multi attribute, but verdicts are not modeled as typed MR verdict records with pass/fail/inconclusive/unsupported semantics.
- **2025 – MT with physical constraints for fluid velocity prediction ML models**
 - Yu et al. propose a **metamorphosis test method integrating physical constraints** for intelligent velocity field prediction [5].
 - **Key points:**
 - Disturbance and cross domain scenarios are designed.
 - MRs are built from **Reynolds number conservation** and **Strouhal number correlations** to test physical consistency and extrapolation [5].
 - Emphasizes **quantitative evaluation** of physical consistency and extrapolation capability.
 - **Progress and remaining gaps:**
 - Clear example of **physics based MRs for ML surrogates in fluid dynamics**—very close thematically to your PDE/CFD surrogate context.
 - However:
 - Regime of validity of these dimensionless–number based MRs is not captured as formal precondition metadata.
 - No explicit error budget/measurement floor analysis; experimental and numerical uncertainties are not used as gating thresholds.
 - Verdicts seem to be scalar performance metrics rather than typed verdict records with explicit reasoning about admissibility.
 - **Trend across [4,5]:**
 - Emergence of **CFD like SciML testing with physical constraint MRs** scenario based evaluation, and emphasis on extrapolation.
 - Still, the *workflow* remains primarily **metric driven evaluation** not the structured MR card plus typed verdict paradigm you seek.

Metamorphic Testing Infrastructure, MR Representation, and Applicability Constraints (2020s)

A separate but highly relevant thread focuses on **general MT methodology, MR representation, and MR applicability**—mostly not domain specific to physics, but foundational for your workflow.

- **2023 – Vision of a “complete MT pipeline”**
 - Duque Torres & Pfahl outline a “complete metamorphic testing pipeline” [2].
 - **Core program:**
 - Automated or semi automated **MR generation** (including a domain specific language for describing MRs).
 - A **constraint definition phase** to capture MR applicability conditions [2].
 - Emphasis on **explainability** of MR outcomes, particularly for regression testing.
 - **Relative to your goals:**
 - Important conceptual advance: explicit recognition that **MRs are not universally applicable** and require **formal constraints**.
 - However, in the available material:
 - Constraints are not framed in terms of physical/nondimensional regimes or measurement floors, but as generic logical conditions over inputs.
 - No explicit integration with scientific diagnostics (residuals, UQ, trust regions, etc.).
 - Verdicts remain essentially binary, although explainability is emphasized.
- **2023 – Interpreting MR violations: bug vs. MR misfit**
 - Duque Torres et al. analyze “Bug or not bug? Analysing the reasons behind metamorphic relation violations” [3].
 - **Contribution:**
 - Uses **Association Rule Mining (ARM)** on test data to infer patterns behind MR violations.
 - Aims to distinguish violations due to true bugs from those due to MRs not fitting specific test data conditions [3].
 - **Relevance to your workflow:**
 - Moves toward a **typed interpretation of MR outcomes**, in the sense of distinguishing root causes (bug vs. MR mis applicability).
 - However:
 -

The classification is an *analysis layer* on top of binary outcomes, not a formally modeled typed verdict in the test artifacts themselves.

- There is no measurement floor concept; MR misfit is inferred statistically, not via explicit admissibility conditions.

- **2024 – Standardized, machine readable MR representation and constrained MRs**

- Duque Torres introduces a framework for “Selecting and Constraining Metamorphic Relations” [1].

- **Key advances:**

- Proposes a **comprehensive repository** and a **standardized, machine readable MR representation**[1].
- Introduces **MetraTrimmer**, a test data driven method to **select and constrain** MRs, emphasizing that MRs are not universally applicable and should be tailored to regions of the test data space [1].

- **Relation to your target:**

- The **MR representation** and **constraint concept** are the closest existing ingredients to what you call MR “cards” and domain validity gating.
- But:
 - Constraints are primarily **empirical/test data driven** rather than physics regime driven.
 - There is no explicit integration of numerical error, measurement noise, or scientific diagnostics in defining constraints or admissibility.
 - Verdicts are still fundamentally pass/fail; the sophisticated part is **which MRs to apply where**, not how verdicts are typed.

- **Trend across [2,3,1]:**

- Development of a **coherent research line** on:
 - Formal MR representation,
 - MR applicability constraints, and
 - Interpretable analysis of MR violations.
- This thread provides key **methodological tools** for your envisioned workflow, but it is not instantiated in a physics/CFD/SciML context and does not incorporate measurement floor admissibility or SciML diagnostics.

Clusters of Contributors and Research Threads

Chen/Tse and early MT for numerical programs [10]

- Early demonstration of MT for PDE programs [10].
- Impact appears foundational but **not extended into a full V&V workflow**; later work on simulation validation and SciML MT is largely independent conceptually.

Olsen & Raunak – MT as simulation validation [8]

- Create a **conceptual bridge** between MT and simulation validation, focusing on ABM and DES [8].
- Their framework and case studies establish MT as a **validation** (not just verification) technique in scientific modeling.
- They do not, however, push into **formal MR artifacts** or **diagnostics gated workflows**

Li et al. – Physics derived MRs in nuclear power software [9]

- Demonstrate **systematic extraction of MRs from physics and numerics** in a safety critical domain.
- Suggests there is appetite in high consequence engineering for MR based V&V, but the treatment of MR validity and error budgets remains informal.

Duque Torres and collaborators – MR representation, constraints, and interpretability [1,2,3]

- Provide the **most coherent sustained thread** on:
 - Machine readable MR representation,
 - Automated constraint definition/selection, and
 - Analysis of MR violations’ causes.
- This work is **methodologically central** for your proposed MR card and constraint notions, but:
 - Is domain agnostic (not physics specific), and

- Does not integrate scientific error budgets or SciML validation tools.

Yang, Chui, Reichert, Shen – MT for hydrologic ML models [6,7]

- Establish MT as a tool for **reliability assessment of ML models** in environmental/hydrological contexts.
- Emphasize qualitative/physics inspired MRs and **extrapolation diagnostics** [7,6].
- Do not formalize MR artifacts or typed verdicts, and do not integrate uncertainty/residual tools as MR gates.

Qi, Yu and collaborators – MT with physical constraints for velocity field prediction [4,5]

- Represent a **newer cluster** focused on:
 - MT for **intelligent physical field prediction**
 - Scenario based frameworks (benchmark/extreme/perturbation/cross domain), and
 - MRs based on **dimensionless quantities** (Re, Strouhal) [4,5].
- Align closely with your PDE/CFD surrogate application, but again **lack formal MR metadata, measurement floor gating, and typed verdicts**.

Emerging Trends and Their Significance

1. From Code Verification toward Model Validation and Surrogate Reliability

- Early work [10] is essentially **code centric** MT checks numerical implementations of PDEs.
- By 2019 [8], MT is explicitly repositioned as a **validation tool for simulations**.
- Recent works [7,6,4,5] extend this to **ML surrogates** in physical domains, focusing on:
 - Reliability without full ground truth,
 - Physical consistency, and
 - Extrapolation beyond training/calibration data.
- **Significance for your contribution:**
 - The community has embraced MT as a way to deal with the **oracle problem in scientific and ML contexts**, but has not yet organized this into a **formal, auditable V&V workflow** with domain gated assets.

2. Growing Recognition of MR Applicability and Constraints

- From [10] through [8,9] and into [7,6,4,5], MR applicability is often discussed qualitatively (e.g., “valid for this model,” “appropriate for this regime”) but not formalized.
- Duque Torres’s line of work [2,3,1] is the first to:
 - Treat MR **constraints/applicability** as first class objects, and
 - Propose a **standardized, machine readable MR representation**[1].
- **Significance:**
 - There is clear awareness that **MRs are not universally valid**, and research is moving toward **formal constraint specification**.
 - However, the constraints are primarily **software/testing space oriented** not *physically or numerically grounded* in nondimensional regimes or error budgets.

3. MT for ML and Scientific Surrogates: Physical Consistency and Extrapolation

- Hydrologic ML works [7,6] and CFD like surrogate works [4,5] share:
 - Use of **physics based or domain knowledge MRs**
 - Focus on **physical consistency** (monotonicity, invariants, physical correlations), and
 - Interest in **extrapolation reliability** (e.g., cross domain scenarios, climate change scenarios).
- **Significance:**
 - These domains are **ripe** for a more principled, MR centric V&V workflow, but currently rely on **ad hoc or qualitative MR specifications and untaped verdicts**.

4. Lack of Measurement Floor Admissibility and Typed Verdicts

Across all the surveyed works:

- **Measurement floor / error budget admissibility:**
 - Tolerances appear as **ad hoc thresholds** (e.g., acceptable error levels) but are not derived from:
 - Discretization error analysis,
 - Experimental uncertainty, or
 - Certified error bounds.
 - No work uses these error budgets to **gate MR evaluation** or to mark tests as “inconclusive” when the measurement floor dominates.
- **Typed verdicts:**
 - All studies essentially operate with **binary MR outcomes** (satisfied/violated) or numerical performance metrics.
 - Even when interpreting violations [3] or combining multiple attributes [4], the **test artifact** does not store structured verdict types such as:
 - Pass,
 - Fail (with satisfied preconditions),
 - Inconclusive (due to high uncertainty or measurement floor),
 - Unsupported (out of domain or MR inapplicable).
- **Significance:**
 - This is a **clear and persistent gap** between current practice and your envisioned domain validity gated workflow.
 - It presents a strong opportunity to introduce **typed verdict records and explicit admissibility logic** as novel contributions.

5. Minimal Integration with SciML Diagnostics as MR Gates

- While several works involve ML models [7,6,4,5], they use:
 - Calibration/sensitivity analyses [6],
 - Scenario based stress tests [4,5],
 - But **do not** integrate:
 - Residual diagnostics from PDE surrogates,
 - UQ or conformal prediction intervals,
 - Trust region/extrapolation indicators, or
 - Certified error bounds as **gating conditions for MR tests or as factors in verdict typing**.
- **Significance:**
 - The **interface between MT and SciML V&V tools** remains largely unexplored.
 - Your proposal to use these diagnostics for **domain validity gating and verdict typing** is not anticipated in the existing literature.

Implications for Positioning a New Domain Validity Gated Workflow

Given this timeline and pattern of developments, a new contribution can be positioned as follows:

1. **Building on established pillars:**
 - **Physics based MRs for scientific codes and surrogates** are well established [10,8,9,7,6,4,5].
 - **MT as a tool for simulation and surrogate validation** (not just verification) is recognized [8,7,6].
 - **Formal MR representation and applicability constraints** have emerging foundations in Duque Torres’s work [2,3,1].
1. **Novel integrative elements not present in prior work:**
 - **Domain validity gated MR cards for physics/CFD/SciML surrogates:**
 - Attach explicit **regime of validity constraints** (nondimensional parameters, mesh/time step ranges, closure models, training data regimes) as machine readable preconditions to each MR.
 - This goes beyond generic constraint fields [1,2] by rooting them in **physics and numerical analysis**.
 - **Measurement floor/admissibility conditions:**
 - Derive **error budgets** from discretization analysis, solver residuals, experimental noise, UQ, or certified bounds, and encode them as part of MR preconditions.
 - Use these budgets to determine when MR evaluation is **meaningful** and when the verdict must be **inconclusive**.

- This is not present in any of [1,2,3,4,5,6,7,8,9,10].
- **Typed verdict records as first class V&V assets:**
 - Formalize verdict types (pass/fail/inconclusive/unsupported) and their evidence (MR residuals, regime checks, diagnostics) in the test data model.
 - This transcends binary outcomes in existing MT work and the post hoc interpretation of violations [3].
- **Integration with SciML diagnostics:**
 - Use residuals, UQ, trust region indicators, conformal intervals, and certified bounds **explicitly to gate MR testing and type verdicts.**
 - This closes the current gap between MT and SciML validation methods.

1. Framing within historical development:

- Your workflow can be positioned as the **next logical step** in the evolution from:
 - Early MT for PDE programs [10] '
 - MT for simulation validation [8] '
 - Physics derived MRs for safety critical codes [9] and ML models [7,6,4,5] '
 - Formal MR representation and constraints [1,2,3] toward a **fully auditable, domain aware MT based V&V framework** for physics simulation surrogates.
- It fills the **structural and methodological gaps** that remain:
 - No existing work combines **physics derived MRs, domain validity gating, measurement floor admissibility, explicit MR cards, and typed verdict records** into a unified, executable workflow.

Overall, the timeline shows a steady maturation of MT in scientific and ML domains, but also a clear absence of the **domain validity gated, error budget aware, typed verdict workflow** you are considering. This provides solid historical grounding and a clear differentiation space for your proposed contribution.

Foundational Work

Which papers form the foundational references on this topic?

The below table shows the resources that are most often cited by the relevant papers on this topic. This is measured by the **reference rate**, which is the fraction of relevant papers that cite a resource. Use this table to determine the most important core papers to be familiar with if you want to deeply understand this topic. Some of these core papers may not be directly relevant to the topic, but provide important context.

Ref.	Reference Rate	Title	Cited By These Relevant Papers
[26]	0.54	Contextual Understanding and Improvement of Metamorphic Testing in Scientific Software Development	[1, 2, 3]
[49]	0.40	Metamorphic Testing: A New Approach for Generating Next Test Cases	[2, 4, 5, 8, 13]
[62]	0.31	A Survey on Metamorphic Testing	[2, 7, 8, 13, 14]
[141]	0.29	Testing and Validating Machine Learning Classifiers by Metamorphic Testing	[4, 6, 7, 8]
[138]	0.22	Metamorphic Testing: Testing the Untestable	[3, 9, 13]
[3]	0.21	Bug or not Bug? Analysing the Reasons Behind Metamorphic Relation Violations	[2]
[18]	0.19	Exploratory Metamorphic Testing for Scientific Software	[2, 7]
[142]	0.18	On Testing Non-Testable Programs	[5, 8, 10, 22]
[110]	0.18	A Replication Study on Predicting Metamorphic Relations at Unit Testing Level	[2]
[28]	0.17	An Application of Metamorphic Testing for Testing Scientific Software	[4, 7, 16, 18, 20]
[143]	0.17	Metamorphic relation prioritization for effective regression testing	[2]
[41]	0.14	An innovative approach for testing bioinformatics programs using metamorphic testing	[7, 8, 14, 22]
[144]	0.14	Metamorphic Testing for Quality Assurance of Protein Function Prediction Tools	[2, 26]

[24]	0.13	Predicting metamorphic relations for testing scientific software: a machine learning approach using graph kernels	[2, 16, 23]
[81]	0.13	Using Semi-Supervised Learning for Predicting Metamorphic Relations	[2, 26]
[108]	0.13	Automated inference of likely metamorphic relations for model transformations	[2]
[145]	0.13	Using machine learning techniques to detect metamorphic relations for programs without test oracles	[2, 22]
[146]	0.13	Quality Assurance of Bioinformatics Software: A Case Study of Testing a Biomedical Text Processing Tool Using Metamorphic Testing	[2, 26]
[147]	0.13	Search-based inference of polynomial metamorphic relations	[2]
[148]	0.12	The Oracle Problem in Software Testing: A Survey	[7, 8, 26]

Adjacent Work

Which papers cite the same foundational papers as relevant papers?

Use this table to discover related papers on adjacent topics, to gain a broader understanding of the field and help generate ideas for useful new research directions.

Ref.	Adjacency score	Title	References These Foundational Papers
[88]	0.06	Variable Discovery with Large Language Models for Metamorphic Testing of Scientific Software	[10, 18, 22, 28, 41, 62, 138]
[62]	0.04	A Survey on Metamorphic Testing	[10, 22, 41]
[7]	0.03	Reliability Assessment of Machine Learning Models in Hydrological Predictions Through Metamorphic Testing	[18, 20, 28, 41, 62]
[26]	0.03	Contextual Understanding and Improvement of Metamorphic Testing in Scientific Software Development	[18, 20, 28, 62]
[160]	0.03	Application of metamorphic testing monitored by test adequacy in a Monte Carlo simulation program	[10, 28, 41]
[83]	0.03	A Study of Oracle Approximations in Testing Deep Learning Libraries	[10, 22, 41]
[18]	0.03	Exploratory Metamorphic Testing for Scientific Software	[10, 20, 28, 49, 62]
[161]	0.03	A mapping study on testing non-testable systems	[10, 22, 41]
[23]	0.03	Testing Scientific Software with Invariant Relations: A Case Study	[10, 28, 49, 62]
[34]	0.03	Testing Ocean Software with Metamorphic Testing	[10, 20, 41, 138]
[8]	0.03	Increasing Validity of Simulation Models Through Metamorphic Testing	[22, 41, 49, 62]
[20]	0.03	Hierarchical Metamorphic Relations for Testing Scientific Software	[10, 28, 49, 62]
[114]	0.03	Unit Tests of Scientific Software: A Study on SWMM	[18, 20, 28, 62]
[67]	0.03	Testing Multiple Linear Regression Systems with Metamorphic Testing	[10, 20, 28, 138]
[99]	0.03	Metamorphic Runtime Checking of Applications without Test Oracles	[22, 41, 145, 157]
[162]	0.02	Multi-objective optimization of cloud systems	[28, 49, 62, 138]
[119]	0.02	The acceptance, use, and perceptions of metamorphic testing for a sample of open-source software developers	[20, 28, 49, 62]
[22]	0.02	Techniques for testing scientific programs without an oracle	[10, 41, 145]
[163]	0.02	Identification Algorithm Framework and Structural model on Input Pattern of Metamorphic Relations	[28, 41]
[164]	0.02	A partial oracle for uniformity statistics	[22, 41]

References

- [1] [Selecting and Constraining Metamorphic Relations](#)
Alejandra Duque-Torres. 2024 IEEE/ACM 46th International Conference on Software Engineering: Companion Proceedings (ICSE-Companion), 2024. 3 citations.
43% Topic Match
Proposes a repository and data driven method to select and constrain metamorphic relations.
Builds a standardized, machine readable MR representation and introduces MetraTrimmer for test data driven MR selection/constraint.
Does not present domain validity gating by physics/regime, measurement floor/error budget admissibility, formal MR cards with preconditions/metrics, typed verdict records, or ties to SciML diagnostics—so relevance to the desired end to end, auditable V&V workflow is limited.
- [2] [Towards a Complete Metamorphic Testing Pipeline](#)
Alejandra Duque-Torres and Dietmar Pfahl. 2023 IEEE International Conference on Software Maintenance and Evolution (ICSME), 2023. 3 citations.
37% Topic Match
Proposes an automated “metamorphic testing pipeline” for MR generation, constraint definition, and explainability.
Describes using automated MR-generation techniques (including a domain-specific language), an MR-constraint phase, and empirical evaluation in regression testing.
Does not present measurement-floor/error-budget admissibility, explicit MR cards/schemas, typed verdict records, or integration with SciML diagnostics—so only partially relevant to your domain-validity-gated workflow.
- [3] [Bug or not Bug? Analysing the Reasons Behind Metamorphic Relation Violations](#)
Alejandra Duque-Torres, ..., and Stefan Fischer. 2023 IEEE International Conference on Software Analysis, Evolution and Reengineering (SANER), 2023. 9 citations.
33% Topic Match
Proposes a method to refine metamorphic relations to diagnose violation causes.
Uses association rule mining (ARM) to extract conditions that explain whether MR violations signal bugs or MR mismatches; validated as a proof-of-concept on a toy example.
Does not appear to provide MR cards, domain-validity gating, quantified measurement-floor/error budgets, or typed (non-binary) verdict artifacts—results are initial and focused on heuristic explanation rather than auditable V&V workflow.
- [4] [Research on Multi-scenario Collaborative Testing Methods for Intelligent Algorithms in Physical Field Prediction](#)
Lianzhi Qi, ..., and Yafen Shen. Proceedings of the 2025 5th International Conference on Computational Modeling, Simulation and Data Analysis, 2025. 0 citations.
30% Topic Match
Proposes a multi-scenario, constraint rule coupled metamorphic testing framework for intelligent physical field prediction.
Implements hierarchical scenarios (benchmark/extreme/perturbation/cross domain), derives MRs from multi parameter physical correlations/statistics, and evaluates via multi attribute scenario-aware metrics on cylinder flow.
Does not appear to formalize domain validity gating via explicit measurement floor/error budgets, MR cards, or typed (pass/fail/inconclusive/unsupported) verdict records—focus is on scenario taxonomy, MR generation, and aggregated evaluation rather than auditable V&V assets.
- [5] [Research on Mutation Testing Strategies for Intelligent Models Predicting Fluid Velocity Fields](#)
Ke Yu, ..., and Lianzhi Qi. 2025 7th International Academic Exchange Conference on Science and Technology Innovation (IAECST), 2025. 0 citations.
26% Topic Match
Proposes a metamorphic-testing method integrating physics-based constraints for ML velocity-field predictors.
Constructs MRs from Reynolds-number conservation and Strouhal-number correlations, using disturbance and cross-domain scenarios to evaluate physical consistency and extrapolation.
Lacks explicit discussion of regime-of-validity gating, quantified measurement-floor/error budgets, formal MR cards, or typed (pass/fail/inconclusive) verdict records—appears methodological/case-study level.
- [6] [Metamorphic testing of machine learning and conceptual hydrologic models](#)
P. Reichert, ..., and Chaopeng Shen. Hydrology and Earth System Sciences, 2024. 18 citations.
22% Topic Match
Applies metamorphic testing to hydrologic conceptual and ML models.
Uses predefined input perturbations, multi-model comparisons, and sensitivity analyses to check qualitative/quantitative consistency.
Does not appear to formalize MR cards, domain-validity gating, measurement-floor error budgets, or typed (non-binary) verdict artifacts; focuses on model comparison and robustness rather than executable, auditable V&V workflow.
- [7] [Reliability Assessment of Machine Learning Models in Hydrological Predictions Through Metamorphic Testing](#)
Yang Yang and T. M. Chui. Water Resources Research, 2021. 18 citations.
20% Topic Match
Applies metamorphic testing to assess ML hydrological predictions without ground truth.
Defines physics-derived MRs (e.g., runoff increases with rainfall) and checks prediction consistency across related inputs.
Lacks explicit domain-validity gating, error-budget/measurement-floor analysis, formal MR cards, or typed (inconclusive/unsupported) verdicts—focuses on empirical MT effectiveness and input similarity.
- [8] [Increasing Validity of Simulation Models Through Metamorphic Testing](#)
Megan M. Olsen and M. Raunak. IEEE Transactions on Reliability, 2019. 27 citations.
18% Topic Match
Proposes applying metamorphic testing to validate executable simulation models.
Presents a framework and guidelines plus three case studies (ABM gossip, ABM cancer, DES airport) to derive pseudo-oracles via MRs.
Does not appear to formalize domain-validity gating, measurement-floor error budgets, explicit MR cards, or typed (inconclusive/unsupported) verdicts—MRs and applicability are discussed informally for ABM/DES contexts.
- [9] [A Lightweight Verification Method Based on Metamorphic Relation for Nuclear Power Software](#)
Meng Li, ..., and Jun Sun. Unknown journal, 2022. 1 citations.
16% Topic Match
Proposes a lightweight metamorphic-testing verification method for nuclear-power PDE software.
Identifies MRs from physics, numerical algorithms, and specifications; applies them to unit/integration/system tests to detect coding/numerical errors without benchmarks.
Does not present formal MR cards, regime-of-validity gating, quantified measurement-floor/error budgets, or typed/inconclusive verdict records; no integration with residual/UQ/trust-region or certified-error tools.
- [10] [Metamorphic testing of programs on partial differential equations: a case study](#)
T. Chen, ..., and T. H. Tse. Proceedings 26th Annual International Computer Software and Applications, 2002. 123 citations.

14% Topic Match

No summary or abstract available

[11] [Metamorphic Testing on the Continuum of Verification and Validation of Simulation Models](#)

M. Raunak and Megan M. Olsen. 2021 IEEE/ACM 6th International Workshop on Metamorphic Testing (MET), 2021. 8 citations.

12% Topic Match

Abstract: Metamorphic testing has been shown to be useful in testing "non-testable" programs in many domains. Modeling & simulation is one such domain, where both verification and validation can be difficult due to lack of oracles. Although the definition of verification and validation vary slightly in modeling and simulation when compared to standard software, we show that metamorphic testing is appropriate in both aspects of ensuring that a simulation model is accurate. In this paper we expand on our five years of prior work on metamorphic testing for simulation validation to show how metamorphic testing can be used for verification as...

[12] [Using Constraint Solvers to Support Metamorphic Testing](#)

M. C. D. Castro-Cabrera, ..., and I. Medina-Bulo. 2019 IEEE/ACM 4th International Workshop on Metamorphic Testing (MET), 2019. 4 citations.

10% Topic Match

Abstract: One of the current challenges in the context of Metamorphic Testing (MT) is the formalization and validation of metamorphic relations (MRs), as there is no single method or homogeneous way of doing it. It is a part of this software testing technique that, unlike others, is not yet developed. On one hand, the fact of having an artifact that formally validates these main elements in MT, facilitates the task for developers and testers and ensures that the technique applied fulfills its function with guarantees. On the other hand, nowadays, there are numerous accessible tools based on highly consolidated and mature...

[13] [MetJam: Metamorphic Testing for Data Synthesize and Quality Assurance of an ML-based Jamming \(DOS\) Detector in 5G](#)

N. B. Chaleshtari, ..., and Andrew Walenstein. 2025 IEEE Canadian Conference on Electrical and Computer Engineering (CCECE), 2025. 0 citations.

9% Topic Match

Abstract: Traditionally, metamorphic testing is an approach for quality assurance engineers to generate test cases with oracles for purposes of assuring the functional correctness of a system under test. We provide an argument for utilizing the metamorphic testing technique for the purpose of domain and application specific data synthesis for machine-learning systems, and we propose a process for creating and maintaining a library of metamorphic relations that concisely captures subject matter expert knowledge about classification boundaries in a way that is easy for them to understand and manage. As is typical in machine-learning applications, synthesized ground truth data of the sort...

[14] [A Tale from the Trenches: Applying Metamorphic and Differential Testing to Bioinformatics Software](#)

Alexis L. Marsh, ..., and R. Cottingham. 2025 IEEE Conference on Software Testing, Verification and Validation (ICST), 2025. 0 citations.

7% Topic Match

Abstract: Metamorphic and differential testing have been proposed as best practices for testing software that is difficult to test, such as for programs in scientific domains. An assumption is that these approaches can be easily customized and applied to almost any domain. However, scientific software is often data-driven, and metamorphic relations may require significant domain knowledge to develop. In addition, tools are often written for ad-hoc experimentation by the scientists and often embed many assumptions about the importance and representation of different natural phenomena. In this paper, we present our experience applying both metamorphic and differential testing to a set of...

[15] [Title Metamorphic testing of programs on partial differentialequations : a case study](#)

T. Y. Chen, ..., and T. H. Tse. Journal Not Provided, 2011. 0 citations.

6% Topic Match

No summary or abstract available

[16] [A Machine Learning Based Framework for Verification and Validation of Massive Scale Image Data](#)

Junhua Ding, ..., and V. Gudivada. IEEE Transactions on Big Data, 2021. 36 citations.

6% Topic Match

Abstract: Big data validation and system verification are crucial for ensuring the quality of big data applications. However, a rigorous technique for such tasks is yet to emerge. During the past decade, we have developed a big data system called CMA for investigating the classification of biological cells based on cell morphology which is captured in diffraction images. CMA includes a collection of scientific software tools, machine learning algorithms, and a large-scale cell image repository. In order to ensure the quality of big data system CMA, we developed a framework for rigorously validating the massive scale image data as well as...

[17] [Application of metamorphic testing in numerical analysis](#)

F. T. Chan, ..., and S. Yiu. Unknown journal, 1998. 61 citations.

5% Topic Match

No summary or abstract available

[18] [Exploratory Metamorphic Testing for Scientific Software](#)

Xuanyi Lin, ..., and D. Rouson. Computing in Science & Engineering, 2018. 29 citations.

5% Topic Match

Abstract: Scientific model developers are able to verify and validate their software via metamorphic testing (MT), even when the expected output of a given test case is not readily available. The tenet is to check whether certain relations hold among the expected outputs of multiple-related inputs. Contemporary approaches require the relations to be defined before tests. Our experience shows that it is often straightforward to first define the multiple iterations of tests for performing continuous simulations, and then keep multiple and even competing metamorphic relations open for investigating the testing-result patterns. We call this new approach exploratory MT, and report our...

[19] [Boosting Metamorphic Testing: A General Metamorphic Specification Language and A Supporting System](#)

Chang-ai Sun, ..., and Pak-Lok Poon. ACM Transactions on Software Engineering and Methodology, 2026. 0 citations.

4% Topic Match

Abstract: Metamorphic testing (MT) is a black-box testing technique to alleviate the oracle problem via leveraging Metamorphic Relations (MRs) based on the domain knowledge of the software under test. In recent years, software testing researchers have made significant research advancements on MT in fundamental theories (e.g., MR identification and composition), methodologies (e.g., test input generation), and fault detection effectiveness in various application domains. However, there still exist some major challenges of MT yet to be addressed, such as the needs for a general MR description language, and an automated system that supports all major steps of MT and integrates the various...

[20] [Hierarchical Metamorphic Relations for Testing Scientific Software](#)

Xuanyi Lin, ..., and Nan Niu. 2018 IEEE/ACM 13th International Workshop on Software Engineering for Science (SE4Science), 2018. 22 citations.

4% Topic Match

Abstract: Scientist developers have not yet routinely adopted systematic testing techniques to assure software quality. A key challenge is the oracle problem, a situation in which appropriate mechanisms are unavailable for checking if the code produces the expected output when executed using a set of test cases (TCs). Metamorphic testing alleviates the oracle problem by specifying the relationship that a source TC and its follow-up TC shall meet. Such relationships are called metamorphic relations (MRs) which are necessary properties of the intended program's functionality. Existing approaches handle the MRs in a flat manner. This paper introduces a novel way to facilitate...

[21] [Metamorphic testing of a Monte Carlo modeling program](#)

Junhua Ding, ..., and Xin-Hua Hu. Unknown journal, 2011. 18 citations.

4% Topic Match

No summary or abstract available

[22] [Techniques for testing scientific programs without an oracle](#)

Upulee Kanewala and J. Bieman. 2013 5th International Workshop on Software Engineering for Computational Science and Engineering (SE-CSE), 2013. 36 citations.

3% Topic Match

No summary or abstract available

[23] [Testing Scientific Software with Invariant Relations: A Case Study](#)

Junhua Ding, ..., and Xin-Hua Hu. 2019 IEEE 19th International Conference on Software Quality, Reliability and Security (QRS), 2019. 4 citations.

3% Topic Match

Abstract: Adequately testing scientific software is essential to the quality of the software. However, it is a grand challenge due to the oracle problem. Metamorphic testing has shown its effectiveness for alleviating the problem. But the effectiveness of metamorphic testing is highly dependent on the quality of metamorphic relations that are developed for testing the software. In this paper, we propose a framework for iteratively developing metamorphic relations for adequately testing scientific software. The basic idea is to refine metamorphic relations that are loosely defined to those that can be verified with only limited number of cases so that the relations...

[24] [Predicting metamorphic relations for testing scientific software: a machine learning approach using graph kernels](#)

Upulee Kanewala, ..., and Asa Ben-Hur. Software Testing, 2016. 113 citations.

3% Topic Match

No summary or abstract available

[25] [A Blueprint for Self-Evolving Coding Agents in Vehicle Aerodynamic Drag Prediction](#)

Jinhui Ren, ..., and Jianmin Wu. ArXiv, 2026. 0 citations.

2% Topic Match

Abstract: High-fidelity vehicle drag evaluation is constrained less by solver runtime than by workflow friction: geometry cleanup, meshing retries, queue contention, and reproducibility failures across teams. We present a contract-centric blueprint for self-evolving coding agents that discover executable surrogate pipelines for predicting drag coefficient C_d under industrial constraints. The method formulates surrogate discovery as constrained optimization over programs, not static model instances, and combines Famou-Agent-style evaluator feedback with population-based island evolution, structured mutations (data, model, loss, and split policies), and multi-objective selection balancing ranking quality, stability, and cost. A hard evaluation contract enforces leakage prevention, deterministic replay, multi-seed robustness, and resource...

[26] [Contextual Understanding and Improvement of Metamorphic Testing in Scientific Software Development](#)

Zedong Peng, ..., and Nan Niu. Proceedings of the 15th ACM / IEEE International Symposium on Empirical Software Engineering and Measurement (ESEM), 2021. 10 citations.

2% Topic Match

Abstract: Background: Metamorphic testing emerges as a simple and effective approach for testing scientific software; yet, its adoption in actual scientific software projects is less studied. Aims: In order for the practitioners to better adopt metamorphic testing in their projects, we set out to first gain a deep understanding about the current quality assurance workflow, testing practices, and tools. Method: We propose to integrate various empirical sources, including artifact analysis, stakeholder interviews, and gap analysis from the literature. Results: Applying our approach to the Open Water Analytics Stormwater Management Model project helped to identify four new needs requiring continued and more...

[27] [Using Metamorphic Relation Violation Regions to Support a Simulation Framework for the Process of Metamorphic Testing](#)

Zhihao Ying, ..., and Z. Zhou. 2022 IEEE 46th Annual Computers, Software, and Applications Conference (COMPSAC), 2022. 6 citations.

2% Topic Match

Abstract: Metamorphic testing (MT) has been growing in popularity, but it can still be quite challenging and time-consuming to assess its performance. Typical approaches to performance assessment can require a series of steps, and depend on a variety of factors, often requiring serendipity. This can be a bottleneck for some aspects of MT research. Central to MT, metamorphic relations (MRs) represent necessary properties of the system under test (SUT). In traditional software testing, simulations are often employed to examine and compare the performance of different testing strategies. However, these simulations are typically designed based on the assumed availability (and applicability) of...

[28] [An Application of Metamorphic Testing for Testing Scientific Software](#)

Junhua Ding, ..., and Xin-Hua Hu. 2016 IEEE/ACM 1st International Workshop on Metamorphic Testing (MET), 2016. 31 citations.

2% Topic Match

No summary or abstract available

[29] [Testing Causality in Scientific Modelling Software](#)

A. Clark, ..., and Robert D. Turner. ACM Transactions on Software Engineering and Methodology, 2022. 17 citations.

2% Topic Match

Abstract: From simulating galaxy formation to viral transmission in a pandemic, scientific models play a pivotal role in developing scientific theories and supporting government policy decisions that affect us all. Given these critical applications, a poor modelling assumption or bug could have far-reaching consequences. However, scientific models possess several properties that make them notoriously difficult to test, including a complex input space, long execution times, and non-determinism, rendering existing testing techniques impractical. In fields such as epidemiology, where researchers seek answers to challenging causal questions, a statistical methodology known as Causal inference has addressed similar problems, enabling the inference of causal...

[30] [Validation-Gated Multi-Agent Governance for Online Adaptation of Thermal-Hydraulic Surrogate Models under Operating-Regime Shift](#)

Doyeong Lim, ..., and In-Cheol Bang. ArXiv, 2026. 0 citations.

2% Topic Match

Abstract: Artificial-intelligence surrogates can support second-by-second thermal-hydraulic forecasting, but models selected and frozen offline may become condition-locked once deployed outside their pretraining envelope. This study develops a guarded continual-adaptation framework for experimental thermal-hydraulic loop data in which role-separated agents - Monitor, Diagnosis, Adaptation, Safety-Auditor, and Orchestrator - diagnose error signatures, prioritize candidate model families, and review promotions, while deterministic champion-challenger gates and background shadow learning retain final authority over model replacement. Seven surrogate families were screened by blocked three-fold cross-validation, and a temporal Fourier neural operator was selected as the initial champion for 60-s-history-to-10-s-trajectory forecasting on two held-out transients, with three...

[31] [Towards Automated Metamorphic Test Identification for Ocean System Models](#)

D. J. Hiremath, ..., and W. Rath. 2021 IEEE/ACM 6th International Workshop on Metamorphic Testing (MET), 2021. 7 citations.

1% Topic Match

Abstract: Metamorphic testing seeks to verify software in the absence of test oracles. Our application domain is ocean system modeling, where test oracles rarely exist, but where symmetries of the simulated physical systems are known. The input data set is large owing to the requirements of the application domain. This paper presents work in progress for the automated generation of metamorphic test scenarios using machine learning. We extended our previously proposed method [1] to identify metamorphic relations with reduced computational complexity. Initially, we represent metamorphic relations as identity maps. We construct a cost function that minimizes for identifying a metamorphic relation orthogonal...

[32] [NOETHER: A Constructive Framework for Metamorphic Pattern Discovery from Operator Algebras](#)

Meng Li, ..., and China. ArXiv, 2026. 0 citations.

1% Topic Match

Abstract: Context. Metamorphic Testing is recognised in IEEE/ISO software-testing standards and increasingly recommended for AI systems, but its progress is bottlenecked by metamorphic relation (MR) identification: existing approaches (structured frameworks, mining and evolutionary pipelines, LLM-assisted methods, MetaPattern catalogues) share an inductive grounding that leaves three foundational questions open: origin, closure, and transferability. Objective. We propose a framework whose downstream step from program-induced operator algebra to MetaPattern set is mechanical and provable, while the upstream curation of the algebra is a stated empirical hypothesis with explicit scope precondition. Method. NOETHER is a two-layer framework. The upstream layer is an eight-block decomposition over...

[33] [Optimization-Embedded Active Multi-Fidelity Surrogate Learning for Multi-Condition Airfoil Shape Optimization](#)

Isaac Robledo, ..., and Rodrigo Castellanos. ArXiv, 2026. 0 citations.

1% Topic Match

Abstract: Active multi-fidelity surrogate modeling is developed for multi-condition airfoil shape optimization to reduce high-fidelity CFD cost while retaining RANS-level accuracy. The framework couples a low-fidelity-informed Gaussian process regression transfer model with uncertainty-triggered sampling and a synchronized elitism rule embedded in a hybrid genetic algorithm. Low-fidelity XFOIL evaluations provide inexpensive features, while sparse RANS simulations are adaptively allocated when predictive uncertainty exceeds a threshold; elite candidates are mandatorily validated at high fidelity, and the population is re-evaluated to prevent evolutionary selection based on outdated fitness values produced by earlier surrogate states. The method is demonstrated for a two-point problem at $Re=6 \times 10^6$...

[34] [Testing Ocean Software with Metamorphic Testing](#)

Quang-Hung Luu, ..., and Hai L. Vu. 2022 IEEE/ACM 7th International Workshop on Metamorphic Testing (MET), 2022. 7 citations.

1% Topic Match

Abstract: Advancing ocean science has a significant impact to the development of the world, from operating a safe navigation for vessels to maintaining a healthy and diverse ocean ecosystem. Various ocean software systems have been extensively adopted for different purposes, for instance, predicting hourly sea level elevation across shorelines, simulating large-scale ocean circulations, as well as integrating into Earth system models for weather forecasts and climate projections. Regardless of their significance, guaranteeing the trustworthiness of ocean software and modelling systems is a long-standing challenge. The testing of ocean software suffers a lot from the so-called oracle problem, which refers to the...

[35] [Towards Generating Executable Metamorphic Relations Using Large Language Models](#)

S. Shin, ..., and Alexandra Baicoianu. Unknown journal, 2024. 12 citations.

1% Topic Match

Abstract: Metamorphic testing (MT) has proven to be a successful solution to automating testing and addressing the oracle problem. However, it entails manually deriving metamorphic relations (MRs) and converting them into an executable form; these steps are time-consuming and may prevent the adoption of MT. In this paper, we propose an approach for automatically deriving executable MRs (EMRs) from requirements using large language models (LLMs). Instead of merely asking the LLM to produce EMRs, our approach relies on a few-shot prompting strategy to instruct the LLM to perform activities in the MT process, by providing requirements and API specifications, as one...

[36] [Testing scientific software: Techniques for automatic detection of metamorphic relations](#)

Upulee Kanewala. Unknown journal, 2015. 0 citations.

1% Topic Match

No summary or abstract available

[37] [Automatically finding Metamorphic Relations in Computational Material Science Parsers](#)

Sebastian Müller, ..., and L. Grunske. 2022 IEEE 18th International Conference on e-Science (e-Science), 2022. 3 citations.

1% Topic Match

Abstract: Software testing is an important part of the software life-cycle. Unfortunately, some software systems have the inherent problem, that it is not clear a priori what the expected outcome is. This is known in the literature as the Oracle Problem. Scientific software suffers from this problem in particular. Metamorphic Testing is a testing approach that mitigates the Oracle Problem, as it is based on identified relations between a program's in- and output pairs. In this study, we investigate the feasibility of automatically finding such metamorphic relations on a software package known as the exciting-NOMAD parser which is widely used in...

[38] [Automated identification of metamorphic test scenarios for an ocean-modeling application](#)

D. J. Hiremath, ..., and W. Rath. 2020 IEEE International Conference On Artificial Intelligence Testing (AITest), 2020. 6 citations.

1% Topic Match

Abstract: Metamorphic testing seeks to validate software in the absence of test oracles. Our application domain is ocean modeling, where test oracles often do not exist, but where symmetries of the simulated physical systems are known. In this short paper we present work in progress for automated generation of metamorphic test scenarios using machine learning. Metamorphic testing may be expressed as $f(g(X))=h(f(X))$ with f being the application under test, with input data X , and with the metamorphic relation (g, h) . Automatically generated metamorphic relations can be used for constructing regression tests, and for comparing different versions of the same software application. Here, we...

[39] [Automated continuous verification for numerical simulation](#)

P. Farrell, ..., and T. Bond. Geoscientific Model Development, 2011. 41 citations.

1% Topic Match

Abstract: Abstract. Verification is a process crucially important for the final users of a computational model: code is useless if its results cannot be relied upon. Typically, verification is seen as a discrete event, performed once and for all after development is complete. However, this does not reflect the reality that many geoscientific codes undergo continuous development of the mathematical model, discretisation and software implementation. Therefore, we advocate that in such cases verification must be continuous and happen in parallel with development: the desirability of their automation follows immediately. This paper discusses a framework for automated continuous verification of wide applicability...

[40] [Testing convolutional neural network based deep learning systems: a statistical metamorphic approach](#)

Faqeer ur Rehman and C. Izurieta. PeerJ Computer Science, 2025. 3 citations.

1% Topic Match

Abstract: Machine learning technology spans many areas and today plays a significant role in addressing a wide range of problems in critical domains, i.e., healthcare, autonomous driving, finance, manufacturing, cybersecurity, etc. Metamorphic testing (MT) is considered a simple but very powerful approach in testing such computationally complex systems for which either an oracle is not available or is available but difficult to apply. Conventional metamorphic testing techniques have certain limitations in verifying deep learning-based models (i.e., convolutional neural networks (CNNs)) that have a stochastic nature (because of randomly initializing the network weights) in their training. In this article, we attempt to...

[41] [An innovative approach for testing bioinformatics programs using metamorphic testing](#)

T. Chen, ..., and Xiaoyuan Xie. BMC Bioinformatics, 2009. 167 citations.

1% Topic Match

Abstract: BackgroundRecent advances in experimental and computational technologies have fueled the development of many sophisticated bioinformatics programs. The correctness of such programs is crucial as incorrectly computed results may lead to wrong biological conclusion or misguided downstream experimentation. Common software testing procedures involve executing the target program with a set of test inputs and then verifying the correctness of the test outputs. However, due to the complexity of many bioinformatics programs, it is often difficult to verify the correctness of the test outputs. Therefore our ability to perform systematic software testing is greatly hindered. ResultsWe propose to use a novel software testing...

[42] [Metamorphic validation for agent-based simulation models](#)

Megan M. Olsen and M. Raunak. Unknown journal, 2016. 12 citations.

1% Topic Match

No summary or abstract available

[43] [Metamorphic Testing for Hybrid Simulation Validation](#)

Mohammed Farhan, ..., and M. Raunak. 2021 Annual Modeling and Simulation Conference (ANNSIM), 2021. 3 citations.

1% Topic Match

Abstract: Proper validation of a simulation model is essential for confidence in its accuracy and credibility. However, many of the most effective approaches for simulation validation require access to data that may be unavailable. Metamorphic Testing (MT), an approach from traditional software testing, has been shown to be useful for verification of simulation software in similar situations. Recent research shows that MT can be applied to the validation of agent based and discrete event simulation models as well. In this paper we build on that work and show how MT can be applied to hybrid simulation models. We demonstrate the effectiveness...

[44] [A method for automated regression test in scientific computing libraries: illustration with SPHinXsys](#)

Bo Zhang, ..., and Xiangyu Hu. ArXiv, 2023. 0 citations.

1% Topic Match

Abstract: Scientific computing libraries, either being in-house or open-source, have experienced enormous progress in both engineering and scientific research. It is therefore essential to ensure that the modifications in the source code aroused by bug fixing or new feature development wouldn't compromise the accuracy and functionality that has already been validated and verified. With this in mind, this paper introduces a method for developing and implementing an automatic regression test environment and takes the open-source multi-physics library SPHInXsys [cite{zhang2021sphinxsys}] as an example. Firstly, the reference database for each benchmark test is generated from monitored data by multiple executions. This database contains...

[45] [Automated engineering of domain-specific metamorphic testing environments](#)

Pablo Gómez-Abajo, ..., and Juan de Lara. Inf. Softw. Technol., 2023. 5 citations.

1% Topic Match

No summary or abstract available

[46] [KLG-MT: a metamorphic testing framework for machine learning-based measuring systems](#)

Paulo Eduardo Eduardo Silva dos Santos, ..., and Wilson de Souza Melo Jr.. Measurement Science and Technology, 2026. 0 citations.

1% Topic Match

Abstract: The increasing adoption of machine learning (ML) in software-controlled measuring systems introduces significant challenges for conformity assessment, particularly under black-box conditions where internal logic and correct outputs cannot be analytically determined. This limitation is closely related to the oracle problem, which restricts the applicability of conventional validation methods based solely on predictive accuracy metrics. To address this challenge, this paper proposes the knowledge-level-guided metamorphic testing (KLG-MT) method, a structured framework that organizes Metamorphic Testing according to progressive levels of operational knowledge available to the tester. The method enables behavioral evaluation of ML-based measuring systems by assessing sensitivity, causal coherence, and...

[47] [Automated regression test method for scientific computing libraries: Illustration with SPHInXsys](#)

Bo Zhang, ..., and Xiangyu Hu. Journal of Hydrodynamics, 2024. 1 citations.

0% Topic Match

No summary or abstract available

[48] [Analyzing the reliability of simulated distributed systems using Metamorphic Testing](#)

A. Núñez, ..., and J. Lara. 2022 IEEE/ACM 7th International Workshop on Metamorphic Testing (MET), 2022. 2 citations.

0% Topic Match

Abstract: Simulation is widely adopted by the research community to analyze and study complex systems. It is based on the idea of creating a model representing the target system under study, so that the experiments can be executed over the model instead of the target system. However, since the model is a simplification of a real-world system, the obtained results entail an accuracy loss, which makes determining the reliability of the experiments a complex task. Testing can be applied to check the correctness of systems. Thus, an oracle is used to determine if a test is correct or not. In the...

[49] [Metamorphic Testing: A New Approach for Generating Next Test Cases](#)

T. Chen, ..., and S. Yiu. ArXiv, 2020. 649 citations.

0% Topic Match

Abstract: In software testing, a set of test cases is constructed according to some predefined selection criteria. The software is then examined against these test cases. Three interesting observations have been made on the current artifacts of software testing. Firstly, an error-revealing test case is considered useful while a successful test case which does not reveal software errors is usually not further investigated. Whether these successful test cases still contain useful information for revealing software errors has not been properly studied. Secondly, no matter how extensive the testing has been conducted in the development phase, errors may still exist in the...

[50] [New ideas: automated engineering of metamorphic testing environments for domain-specific languages](#)

P. Cañizares, ..., and J. Lara. Proceedings of the 14th ACM SIGPLAN International Conference on Software Language Engineering, 2021. 10 citations.

0% Topic Match

Abstract: Two crucial aspects for the trustworthy utilization of domain-specific languages (DSLs) are their semantic correctness, and proper testing support for their users. Testing is frequently used to verify correctness, but is often done informally -- which may yield unreliable results -- and requires substantial effort for creating suitable test cases and oracles. To alleviate this situation, we propose an automated technique for building metamorphic testing environments for DSLs. Metamorphic testing identifies expected relationships between the outputs of two consecutive tests, reducing the effort in specifying oracles and creating test cases manually. This new ideas paper presents the overarching concepts, the...

[51] [Test Adequacy Criteria for Metamorphic Testing](#)

Yanwen Liu, ..., and Zheng Zheng. 2024 IEEE 24th International Conference on Software Quality, Reliability, and Security Companion (QRS-C), 2024. 1 citations.

0% Topic Match

Abstract: The Oracle problem poses a fundamental challenge in software testing, seeking to validate the accuracy of a program's output given specific test inputs. Metamorphic testing (MT) has emerged as an automated technique to tackle the Oracle problem by testing programs against metamorphic relations (MRs) - crucial properties that persist across multiple executions of the target program. An essential aspect of software testing involves defining objective metrics to gauge the violation criteria. While many studies concentrate on the defect monitoring capability of MRs, the adequacy criteria for MT often lack coverage information regarding MRs. In this paper, we proposed MTcoverage, an...

[52] [Model Assisted Refinement of Metamorphic Relations for Scientific Software](#)

Clay Stevens, ..., and Myra B. Cohen. 2025 IEEE/ACM 47th International Conference on Software Engineering: New Ideas and Emerging Results (ICSE-NIER), 2025. 0 citations.

0% Topic Match

Abstract: Ensuring the correctness of scientific software is challenging due to the need to represent and model complex phenomenon in a discrete form. Many dynamic approaches for correctness have been developed for numerical overflow or imprecision, which may manifest as program crashes or hangs. Less effort has been spent on functional correctness, where one of the most widely proposed technique is metamorphic testing. Metamorphic testing often requires deep domain expertise to design meaningful relations. In this vision paper we ask if we can utilize the process of abstraction and refinement, a traditionally formal approach, to guide the development of metamorphic relations....

[53] [Verification and validation for trustworthy scientific machine learning](#)

J. Jakeman, ..., and Thomas O'Leary-Roseberry. Machine Learning: Science and Technology, 2025. 5 citations.

0% Topic Match

Abstract: Scientific machine learning (SciML) models are transforming many scientific disciplines. However, the development of good modeling practices to increase the trustworthiness of SciML has lagged behind its application, limiting its potential impact. The goal of this paper is to start a discussion on establishing consensus-based good practices for predictive SciML. We identify key challenges in applying existing computational science and engineering guidelines, such as verification and validation protocols, and provide recommendations to address these challenges. Our discussion focuses on predictive SciML, which uses machine learning models to learn, improve, and accelerate numerical simulations of physical systems. While centered on predictive...

[54] [Scientific Software Testing Goes Serverless: Creating and Invoking Metamorphic Functions](#)

Xuanyi Lin, ..., and Nan Niu. IEEE Software, 2020. 13 citations.

0% Topic Match

Abstract: Our function-as-a-service (FaaS) framework transformed end users' questions into automated tests for scientific software. Our case study illustrates the FaaSification of scientific software testing and the importance of value-driven evaluations by focusing on real-world defect detection.

[55] [AgentAssay: Token-Efficient Regression Testing for Non-Deterministic AI Agent Workflows](#)

Varun Pratap Bhardwaj. ArXiv, 2026. 4 citations.

0% Topic Match

Abstract: Autonomous AI agents are deployed at unprecedented scale, yet no principled methodology exists for verifying that an agent has not regressed after changes to its prompts, tools, models, or orchestration logic. We present AgentAssay, the first token-efficient framework for regression testing non-deterministic AI agent workflows, achieving 78-100% cost reduction while maintaining rigorous statistical guarantees. Our contributions include: (1) stochastic three-valued verdicts (PASS/FAIL/INCONCLUSIVE) grounded in hypothesis testing; (2) five-dimensional agent coverage metrics; (3) agent-specific mutation testing operators; (4) metamorphic relations for agent workflows; (5) CI/CD deployment gates as statistical decision procedures; (6) behavioral fingerprinting that maps execution traces to compact vectors,...

[56] [Performance Metamorphic Testing: Motivation and Challenges](#)

Sergio Segura, ..., and Antonio Ruiz-Cortés. 2017 IEEE/ACM 39th International Conference on Software Engineering: New Ideas and Emerging Technologies Results Track (ICSE-NIER), 2017. 29 citations.

0% Topic Match

Abstract: Performance testing is a challenging task mainly due to the lack of test oracles, that is, mechanisms to decide whether the performance of a program under a certain workload is either acceptable or poor due to a performance bug. Metamorphic testing enables the generation of test cases in the absence of an oracle by exploiting the relations (so-called metamorphic relations) between the inputs and outputs of multiple executions of the program under test. In the last two decades, metamorphic testing has been successfully used to detect functional faults in a variety of domains, ranging from web services to simulators. However,...

[57] [XRePIT: A deep learning-computational fluid dynamics hybrid framework implemented in OpenFOAM for fast, robust, and scalable unsteady simulations](#)

Shilaj Baral, ..., and Joongoo Jeon. ArXiv, 2025. 1 citations.

0% Topic Match

Abstract: Autoregressive neural surrogates offer computational acceleration for fluid dynamics but inherently suffer from error accumulation and non-physical drift during long-term rollouts. Although hybrid strategies combining surrogate models and physics-based solvers have been proposed, they are limited to manual implementations for low-dimensional benchmarks. In this study, we propose an OpenFOAM-based hybrid framework, XRePIT (eXtensible Residual-based Physics-informed Transfer learning), characterized by its fastness, robustness, and scalability. Unlike prior manual implementations (e.g., RePIT), XRePIT integrates a fully automated open-source workflow that manages the state transition between a neural surrogate and a traditional numerical solver (OpenFOAM) based on a monitored residual threshold. Using 3D...

[58] [Metamorphic relation automation: Rationale, challenges, and solution directions](#)

Emran Altamimi, ..., and C. Catal. Journal of Software: Evolution and Process, 2022. 4 citations.

0% Topic Match

Abstract: Metamorphic testing addresses the issue of the oracle problem by comparing results transformation from multiple test executions. The relationship that governs the output transformation is called metamorphic relation. Metamorphic relations require expert knowledge and the generation of them is considered a time consuming task. Researchers have proposed various techniques to automate metamorphic testing, generation, and selection. Although there are several research articles on this issue, there is a lack of overview of the state of the art of metamorphic relation automation. As such, we performed a systematic literature review study to collect, extract, and synthesize the required data. Based on our research questions, the...

[59] [Finding Metamorphic Relations for Scientific Software](#)

Xuanyi Lin, ..., and Hui Liu. 2021 IEEE/ACM 43rd International Conference on Software Engineering: Companion Proceedings (ICSE-Companion), 2021. 7 citations.

0% Topic Match

Abstract: Metamorphic testing uncovers defects by checking whether a relation holds among multiple software executions. These relations are known as metamorphic relations (MRs). For scientific software operating in a large multi-parameter input space, identifying MRs that determine the simultaneous changes among multiple variables is challenging. In this poster, we propose a fully automatic approach to classifying input and output variables from scientific software's user manual, mining these variables' associations from the user forum to generate MRs, and validating the MRs with existing regression tests. Preliminary results of our end-to-end MR support for the Storm Water Management Model (SWMM) are reported.

[60] [Metamorphic Testing for Vision-Based Autonomous Driving With Road Traffic Risk Exposure Extrapolation](#)

Zhengmin Jiang, ..., and Jianping Wang. IEEE Transactions on Intelligent Transportation Systems, 2026. 0 citations.

0% Topic Match

Abstract: Autonomous Driving Systems (ADS) are critical components of Intelligent Transportation Systems (ITS), where vehicle-level reliability has a direct bearing on road traffic safety. Evaluating ADS performance in complex environments remains challenging due to the absence of test oracles and the heavy reliance on deep learning. To address these challenges, this study proposes a novel metamorphic testing framework tailored for vision-based ADS. First, causal inference is employed to extract key environmental factors from high-dimensional observational traffic data, thereby reducing the test space. Second, a multi-objective optimization algorithm integrating causal counterfactual reasoning is developed to quantify the challenges associated with specific combinations...

[61] [Generative Surrogates in Ansys: Reliability-Driven Multiphysics Optimization](#)

Murali Krishna Pasupuleti. International Journal of Academic and Industrial Research Innovations(IJAIRI), 2023. 0 citations.

0% Topic Match

Abstract: Abstract: Designing multiphysics systems under uncertainty is computationally prohibitive when reliability must be guaranteed at design time. This study proposes a reliability-driven optimization framework that couples Ansys high-fidelity solvers with generative surrogate models to reduce cost while maintaining formal safety margins. The approach represents solver response surfaces using variational autoencoders, diffusion models, and normalizing flows conditioned on design variables and physics priors; reliability constraints are enforced through FORM/SORM, Monte-Carlo estimators, and CVaR-aware acquisition within Bayesian and NSGA-II optimization loops. Methods include uncertainty-calibrated training with conformal risk control, out-of-distribution detection for surrogate gating, and selective high-fidelity verifications to bound model bias;...

[62] [A Survey on Metamorphic Testing](#)

Sergio Segura, ..., and Antonio Ruiz-Cortés. IEEE Transactions on Software Engineering, 2016. 532 citations.

0% Topic Match

Abstract: A test oracle determines whether a test execution reveals a fault, often by comparing the observed program output to the expected output. This is not always practical, for example when a program's input-output relation is complex and difficult to capture formally. Metamorphic testing provides an alternative, where correctness is not determined by checking an individual concrete output, but by applying a transformation to a test input and observing how the program output "morphs" into a different one as a result. Since the introduction of such metamorphic relations in 1998, many contributions on metamorphic testing have been made, and the technique...

[63] [Uncertainty quantification of surrogate models using conformal prediction](#)

Vignesh Gopakumar, ..., and M. Deisenroth. Machine Learning: Science and Technology, 2024. 12 citations.

0% Topic Match

Abstract: Data-driven surrogate models offer fast, inexpensive approximations to complex numerical and experimental systems but typically lack uncertainty quantification, limiting their reliability in safety-critical applications. While Bayesian methods provide uncertainty estimates, they offer no statistical guarantees and struggle with high-dimensional spatio-temporal problems due to computational costs and dependence on prior specification. We present a conformal prediction (CP) framework that provides statistically guaranteed marginal coverage for surrogate models in a model-agnostic manner with near-zero computational cost. Our approach handles high-dimensional spatio-temporal outputs by performing cell-wise calibration while preserving the tensorial structure of predictions. Through extensive empirical evaluation across diverse applications-including partial differential...

[64] [Testing Constraint Checking Implementations via Principled Metamorphic Transformations](#)

Mingchen Gao, ..., and Chang Xu. 2024 IEEE International Conference on Software Analysis, Evolution and Reengineering (SANER), 2024. 1 citations.

0% Topic Match

Abstract: Constraint checking techniques are being widely used for ensuring the consistency of software artifacts during their development and evolution (e.g., detecting inconsistency in an application's running contexts or identifying rule violation in the code being developed). Typically, consistency constraints are formulated and checked upon the changes of software artifacts under checking. When any constraint is violated, an inconsistency is said to occur and then follow-up actions

can be taken to remedy the problem. Currently, various constraint checking techniques have been proposed and implemented with sophisticated mechanisms for higher efficiency and scalability. However, these implementations could be far from being well...

[65] [Discovering Metamorphic Relations for Scientific Software From User Forums](#)

Xuanyi Lin, ..., and Karla Morris. Computing in Science & Engineering, 2021. 5 citations.

0% Topic Match

Abstract: Scientific software can be used for decades and is constantly evolving. Recently, metamorphic testing, a property-based testing technique, has shown to be effective in testing scientific software, and the necessary properties are expressed as metamorphic relations. However, the development of metamorphic relations is difficult: it requires considerable practical expertise for the software tester. In this article, we report our experience of uncovering metamorphic relations from a user forum's questions of the United States Environmental Protection Agency's Storm Water Management Model (SWMM). Our study not only illustrates a wealth of end users' expertise in interpreting software results, but also demonstrates the...

[66] [Metamorphic Testing of a Deep Learning Based Forecaster](#)

Anurag Dwarakanath, ..., and M. Koushik. 2019 IEEE/ACM 4th International Workshop on Metamorphic Testing (MET), 2019. 12 citations.

0% Topic Match

Abstract: In this paper, we present the Metamorphic Testing of an in-use deep learning based forecasting application. The application looks at the past data of system characteristics (e.g. 'memory allocation') to predict outages in the future. We focus on two statistical / machine learning based components - a) detection of co-relation between system characteristics and b) estimating the future value of a system characteristic using an LSTM (a deep learning architecture). In total, 19 Metamorphic Relations have been developed and we provide proofs & algorithms where applicable. We evaluated our method through two settings. In the first, we executed the relations...

[67] [Testing Multiple Linear Regression Systems with Metamorphic Testing](#)

Quang-Hung Luu, ..., and T. Chen. ArXiv, 2021. 31 citations.

0% Topic Match

Abstract: Regression is one of the most commonly used statistical techniques. However, testing regression systems is a great challenge because of the absence of test oracle in general. In this paper, we show that Metamorphic Testing is an effective approach to test multiple linear regression systems. In doing so, we identify intrinsic mathematical properties of linear regression, and then propose 11 Metamorphic Relations to be used for testing. Their effectiveness is examined using mutation analysis with a range of different regression programs. We further look at how the testing could be adopted in a more effective way. Our work is applicable...

[68] [Metamorphic Testing Using Geometric Interrogation Technique and Its Application](#)

K. Sim, ..., and C. Lin. Unknown journal, 1970. 16 citations.

0% Topic Match

Abstract: Usual techniques used in the automatic test case generation and executions assume that a complete oracle will be available during the testing process. However, in some numerical programs this assumption is not valid because the oracle in form of analytical solution is either not available or too difficult to obtain. To overcome this problem, metamorphic testing proposes to generate follow-up test cases to check important properties of the output function. We study the intrinsic properties of Medial Axes for casting applications. Even though Medial Axes is a geometric function that is independent from the output function of casting simulation, we...

[69] [Robust Solution Verification Experiments on Nonuniform Meshes](#)

Justin R. Weinmeister and D. Sanjaya. Volume 1: Artificial Intelligence (AI) for Fluids; CFD Methods; CFD Applications; Bio-Inspired and Biomedical Fluid Dynamics; Fluid Measurement and Instrumentation; Energy & Sustainability, 2025. 0 citations.

0% Topic Match

Abstract: The activities of verification, validation, and uncertainty quantification (VVUQ) provide a comprehensive means to assess the credibility of computational models. Within VVUQ, solution verification assesses numerical errors and evaluates whether the simulation is sufficiently accurate for its intended applications. As computational modeling gains traction in the development of complex, high-consequence systems, the need for robust solution verification intensifies, particularly because experimental data for these systems are often limited. This work examines improvements in the robustness of Richardson extrapolation (RE), a method commonly used in solution verification to study the discretization error of computational models using a power law. Nonuniform...

[70] [Metamorphic Testing for Sequential Prediction Models: A Survey of LSTMs and LLMs](#)

Alejandra Duque-Torres, ..., and Claus Klammer. 2026 IEEE International Conference on Software Analysis, Evolution and Reengineering - Companion (SANER-C), 2026. 0 citations.

0% Topic Match

Abstract: Recurrent neural networks such as Long ShortTerm Memory (LSTMs) and Transformer-based Large Language Models (LLMs) are increasingly deployed in applications that rely on sequential predictions, including time-series forecasting, code generation, dialogue systems, and decision support. Ensuring their reliability, robustness, and fairness remains challenging due to the oracle problem, non-determinism, distribution shift, and rapid model evolution. Metamorphic Testing (MT) has emerged as a promising strategy to alleviate the oracle problem by specifying expected relations between multiple executions rather than absolute outputs. However, while a growing effort applies MT to LSTMs and LLMs, the landscape is fragmented across domains, targeted properties, and...

[71] [A Genetic Algorithm-Based Approach for Composite Metamorphic Relations Construction](#)

Zhenglong Xiang, ..., and Fei Yu. Inf., 2019. 12 citations.

0% Topic Match

Abstract: The test oracle problem exists widely in modern complex software testing, and metamorphic testing (MT) has become a promising testing technique to alleviate this problem. The inference of efficient metamorphic relations (MRs) is the core problem of metamorphic testing. Studies have proven that the combination of simple metamorphic relations can construct more efficient metamorphic relations. In most previous studies, metamorphic relations have been mainly manually inferred by experts with professional knowledge, which is an inefficient technique and hinders the application. In this paper, a genetic algorithm-based approach is proposed to construct composite metamorphic relations automatically for the program to be...

[72] [Metamorphic Testing: Beyond Testing Numerical Computations](#)

Z. Zhou. 2017 IEEE/ACM 2nd International Workshop on Metamorphic Testing (MET), 2017. 1 citations.

0% Topic Match

No summary or abstract available

[73] [Large Language Models: The Next Frontier for Variable Discovery within Metamorphic Testing?](#)

Christos Tsigkanos, ..., and Timo Kehrer. 2023 IEEE International Conference on Software Analysis, Evolution and Reengineering (SANER), 2023. 22 citations.

0% Topic Match

Abstract: Metamorphic testing involves reasoning on necessary properties that a program under test should exhibit regarding multiple input and output variables. A general approach consists of extracting metamorphic relations from auxiliary artifacts such as user manuals or documentation, a strategy particularly fitting to testing scientific software. However, such software typically has large input-output spaces, and the fundamental prerequisite – extracting variables of interest – is an arduous and non-scalable process when performed manually. To this end, we devise a workflow around an autoregressive transformer-based Large Language Model (LLM) towards the extraction of variables from user manuals of scientific software. Our end-to-end...

[74] [Using Metamorphic Relations to Improve The Effectiveness of Automatically Generated Test Cases](#)

Prashanta Saha and Upulee Kanewala. 2022 IEEE/ACIS 20th International Conference on Software Engineering Research, Management and Applications (SERA), 2022. 0 citations.

0% Topic Match

Abstract: Automated test case generation has helped to reduce the cost of testing. However, developing effective test oracles for these automatically generated test cases still remains a challenge. Metamorphic testing (MT) has become a well-known software testing approach over the years. This testing technique can effectively alleviate the oracle problem faced when testing using metamorphic relations (MRs) to determine whether a test case is passed or failed. In this work, we conduct an empirical study on an open source linear algebra library to evaluate whether MRs can be utilized to improve the fault detection effectiveness of automatically generated test cases. Our...

[75] [Parsimonious Uncertainty Quantification for Modeling and Simulation of Flow-Induced Vibrations](#)

Gregory Banyay and A. Nejad. Volume 4: Fluid-Structure Interaction; High Pressure Technology, 2025. 0 citations.

0% Topic Match

Abstract: Uncertainties pervade the process of constructing credible computational models, generating simulation results, and interpretation thereof. One must compare simulation results with experiment results, which intend to represent the real world environment (e.g., model validation via correspondence with observations). One must additionally compare the conceptual and mathematical models by which we represent the real world and how those models become encoded on a computer (e.g., model verification via consistency with first principles). Those evaluations of real-world correspondence and phenomenological consistency represent two necessary and distinct approaches to uncertainty quantification. Systems, structures, and components subject to stochastic dynamics, such as flow-induced...

[76] [Improving Deep Learning Framework Testing with Model-Level Metamorphic Testing](#)

Yanzhou Mu, ..., and Zhenyu Chen. Proceedings of the ACM on Software Engineering, 2025. 7 citations.

0% Topic Match

Abstract: Deep learning (DL) frameworks are essential to DL-based software systems, and framework bugs may lead to substantial disasters, thus requiring effective testing. Researchers adopt DL models or single interfaces as test inputs and analyze their execution results to detect bugs. However, floating-point errors, inherent randomness, and the complexity of test inputs make it challenging to analyze execution results effectively, leading to existing methods suffering from a lack of suitable test oracles. Some researchers utilize metamorphic testing to tackle this challenge. They design Metamorphic Relations (MRs) based on input data and parameter settings of a single framework interface to generate equivalent...

[77] [Calibrated Physics-Informed Uncertainty Quantification](#)

Vignesh Gopakumar, ..., and M. Deisenroth. ArXiv, 2025. 8 citations.

0% Topic Match

Abstract: Simulating complex physical systems is crucial for understanding and predicting phenomena across diverse fields, such as fluid dynamics and heat transfer, as well as plasma physics and structural mechanics. Traditional approaches rely on solving partial differential equations (PDEs) using numerical methods, which are computationally expensive and often prohibitively slow for real-time applications or large-scale simulations. Neural PDEs have emerged as efficient alternatives to these costly numerical solvers, offering significant computational speed-ups. However, their lack of robust uncertainty quantification (UQ) limits deployment in critical applications. We introduce a model-agnostic, physics-informed conformal prediction (CP) framework that provides guaranteed uncertainty estimates without requiring...

[78] [A study of the interactive role of metamorphic testing and machine learning in the quality assurance of a deep learning forecasting application](#)

Islam Nasr, ..., and F. Karray. International Journal of Information Technology, 2023. 18 citations.

0% Topic Match

No summary or abstract available

[79] [Paving the way for scientific foundation models: enhancing generalization and robustness in PDEs with constraint-aware pre-training](#)

Amin Totounferoush, ..., and S. Staab. ArXiv, 2025. 4 citations.

0% Topic Match

Abstract: Partial differential equations (PDEs) govern a wide range of physical systems, but solving them efficiently remains a major challenge. The idea of a scientific foundation model (SciFM) is emerging as a promising tool for learning transferable representations across diverse domains. However, SciFMs require large amounts of solution data, which may be scarce or computationally expensive to generate. To maximize generalization while reducing data dependence, we propose incorporating PDE residuals into pre-training either as the sole learning signal or in combination with data loss to compensate for limited or infeasible training data. We evaluate this constraint-aware pre-training across three key benchmarks:...

[80] [Gotten: A Model-Driven Solution to Engineer Domain-specific Metamorphic Testing Environments](#)

Pablo Gómez-Abajo, ..., and Juan de Lara. 2023 ACM/IEEE International Conference on Model Driven Engineering Languages and Systems Companion (MODELS-C), 2023. 4 citations.

0% Topic Match

Abstract: Testing is essential for assessing the correctness of software systems. Metamorphic testing (MT) is an approach especially suited when the system under test lacks oracles, or they are expensive to compute. However, creating an MT environment for a specific domain (e.g., cloud simulation, model transformation, machine learning) requires substantial effort. To alleviate these difficulties, we present a model-driven tool that automates the construction of MT environments. Starting from a meta-model with the domain concepts, and a description of the domain execution environment, our tool produces an MT environment featuring comprehensive support for the MT process. This includes the definition of...

[81] [Using Semi-Supervised Learning for Predicting Metamorphic Relations](#)

B. Hardin and Upulee Kanewala. 2018 IEEE/ACM 3rd International Workshop on Metamorphic Testing (MET), 2018. 28 citations.

0% Topic Match

Abstract: Software testing is difficult to automate, especially in programs which have no oracle, or method of determining which output is correct. Metamorphic testing is a solution this problem. Metamorphic testing uses metamorphic relations to define test cases and expected outputs. A large amount of time is needed for a domain expert to determine which metamorphic relations can be used to test a given program. Metamorphic relation prediction removes this need for such an expert. We propose a method using semi-supervised machine learning to detect which metamorphic relations are applicable to a given code base. We compare this semi-supervised model with...

[82] [Out-of-distribution generalization of deep-learning surrogates for 2D PDE-generated dynamics in the small-data regime](#)

Binh Duong Nguyen and Stefan Sandfeld. ArXiv, 2026. 2 citations.

0% Topic Match

Abstract: Partial differential equations (PDEs) are a central tool for modeling the dynamics of physical, engineering, and materials systems, but high-fidelity simulations are often computationally expensive. At the same time, many scientific applications can be viewed as the evolution of spatially distributed fields, making data-driven forecasting of such fields a core task in scientific machine learning. In this work we study autoregressive deep-learning surrogates for two-dimensional PDE dynamics on periodic domains, focusing on generalization to out-of-distribution initial conditions within a fixed PDE and parameter regime and on strict small-data settings with at most 10^2 simulated trajectories per system. We introduce a...

[83] [A Study of Oracle Approximations in Testing Deep Learning Libraries](#)

Mahdi Nejadgholi and Jinqiu Yang. 2019 34th IEEE/ACM International Conference on Automated Software Engineering (ASE), 2019. 55 citations.

0% Topic Match

Abstract: Due to the increasing popularity of deep learning (DL) applications, testing DL libraries is becoming more and more important. Different from testing general software, for which output is often asserted definitely (e.g., an output is compared with an oracle for equality), testing deep learning libraries often requires to perform oracle approximations, i.e., the output is allowed to be within a restricted range of the oracle. However, oracle approximation practices have not been studied in prior empirical work that focuses on traditional testing practices. The prevalence, common practices, maintenance and evolution challenges of oracle approximations remain unknown in literature. In this...

[84] [Validating Unsupervised Machine Learning Techniques for Software Defect Prediction With Generic Metamorphic Testing](#)

Pak Yuen Patrick Chan and J. Keung. IEEE Access, 2024. 3 citations.

0% Topic Match

Abstract: In the realm of software defect prediction, unsupervised models often step in when labelled datasets are scarce, despite facing the challenge of validating models without prior knowledge of data. Addressing this, we proposed a novel approach leveraging generic metamorphic testing to validate such models effectively, bypassing the need for expert-derived metamorphic relations. Our method identifies five categories of generic metamorphic relations, further divided into twenty-one individual generic metamorphic relations, all formulated through generic Data Mutation Operators. This framework enables us to generate follow-up datasets from the source datasets, training respective software defect prediction models. By comparing predictions between the source...

[85] [An uncertainty-quantification framework for assessing accuracy, sensitivity, and robustness in computational fluid dynamics](#)

S. Rezaeiravesh, ..., and P. Schlatter. J. Comput. Sci., 2020. 30 citations.

0% Topic Match

Abstract: A framework is developed based on different uncertainty quantification (UQ) techniques in order to assess validation and verification (V&V) metrics in computational physics problems, in general, and computational fluid dynamics (CFD), in particular. The metrics include accuracy, sensitivity and robustness of the simulator's outputs with respect to uncertain inputs and computational parameters. These parameters are divided into two groups: based on the variation of the first group, a computer experiment is designed, the data of which may become uncertain due to the parameters of the second group. To construct a surrogate model based on uncertain data, Gaussian process regression (GPR)...

[86] [Towards Physics Constrained Deep Learning Based Turbulence Model Uncertainty Quantification](#)

Minghan Chu and Weicheng Qian. ArXiv, 2025. 0 citations.

0% Topic Match

Abstract: Turbulence Models represent the workhorse for simulations used in engineering design and analysis. Despite their low computational cost and robustness, these models suffer from substantial predictive uncertainty, most of which is epistemic. At present, the Eigenspace Perturbation Method (EPM) is the only approach to estimate these turbulence model uncertainties, using physics based perturbation to the predicted Reynolds stresses. While the EPM address the question of how to perturb the Reynolds stresses for uncertainty estimation, it does not address how much to perturb. This shortcoming leads to very generous uncertainty bounds that result in sub-optimal designs. In this investigation, we use...

[87] [Semantic Invariance in Agentic AI](#)

I. D. Zarzà, ..., and C. Calafate. ArXiv, 2026. 2 citations.

0% Topic Match

Abstract: Large Language Models (LLMs) increasingly serve as autonomous reasoning agents in decision support, scientific problem-solving, and multi-agent coordination systems. However, deploying LLM agents in consequential applications requires assurance that their reasoning remains stable under semantically equivalent input variations, a property we term semantic invariance. Standard benchmark evaluations, which assess accuracy on fixed, canonical problem formulations, fail to capture this critical reliability dimension. To address this shortcoming, in this paper we present a metamorphic testing framework for systematically assessing the robustness of LLM reasoning agents, applying eight semantic-preserving transformations (identity, paraphrase, fact reordering, expansion, contraction, academic context, business context, and contrastive...

[88] [Variable Discovery with Large Language Models for Metamorphic Testing of Scientific Software](#)

Christos Tsigkanos, ..., and Timo Kehler. Unknown journal, 2023. 16 citations.

0% Topic Match

No summary or abstract available

[89] [Physics Constrained Deep Learning For Turbulence Model Uncertainty Quantification](#)

Minghan Chu and Weicheng Qian. ArXiv, 2024. 1 citations.

0% Topic Match

Abstract: Engineering design and scientific analysis rely upon computer simulations of turbulent fluid flows using turbulence models. These turbulence models are empirical and approximate, leading to large uncertainties in their predictions that hamper scientific and engineering advances. We outline a Physics Constrained Deep Learning framework to estimate turbulence model uncertainties using physics based Eigenspace Perturbations along with Deep Learning based guidance. The Deep Learning based modulation controls the spatial variation in perturbation magnitude to improve the calibration of uncertainty estimates over the state of the art physics based methods.

[90] [Metamorphic Testing: A Simple Yet Effective Approach for Testing Scientific Software](#)

Upulee Kanewala and T. Chen. Computing in Science & Engineering, 2019. 15 citations.

0% Topic Match

Abstract: Testing scientific software is a difficult task due to their inherent complexity and the lack of test oracles. In addition, these software systems are usually developed by end-user developers who are not normally trained as professional software developers nor testers. These factors often lead to inadequate testing. Metamorphic testing (MT) is a simple yet effective testing technique for testing such applications. Even though MT is a wellknown technique in the software testing community, it is not very well utilized by the scientific software developers. The objective of this paper is to present MT as an effective technique for testing scientific...

[91] [Model-Based Metamorphic Testing: A Case Study](#)

Jinhua Ding and Dianxiang Xu. Unknown journal, 2012. 2 citations.

0% Topic Match

No summary or abstract available

[92] [Metamorphic Testing for Semantic Invariance in Large Language Models](#)

J. Curtò and I. D. Zarzà. IEEE Access, 2025. 6 citations.

0% Topic Match

Abstract: Large Language Models (LLMs) have demonstrated remarkable capabilities in mathematical and scientific reasoning tasks, yet standard benchmark evaluations fail to capture the stability of these reasoning processes under semantically equivalent input variations. We introduce a metamorphic testing framework for assessing semantic invariance in LLM reasoning, systematically evaluating model responses across eight metamorphic relations: identity, paraphrase, fact reordering, expansion, contraction, academic context, business context, and contrastive formulation. We conduct a comparative analysis of two recent foundation models—Hermes-4-70B and DeepSeek-R1-0528—across 79 reasoning problems spanning eight categories (Physics, Mathematics, Chemistry, Economics, Statistics, Biology, Calculus, and Optimization) at three difficulty levels. Both models exhibit...

[93] [Physics-Guided Machine Learning for Uncertainty Quantification in Turbulence Models](#)

Minghan Chu and Weicheng Qian. ArXiv, 2025. 0 citations.

0% Topic Match

Abstract: Predicting the evolution of turbulent flows is central across science and engineering. Most studies rely on simulations with turbulence models, whose empirical simplifications introduce epistemic uncertainty. The Eigenspace Perturbation Method (EPM) is a widely used physics-based approach to quantify model-form uncertainty, but being purely physics-based it can overpredict uncertainty bounds. We propose a convolutional neural network (CNN)-based modulation of EPM perturbation magnitudes to improve calibration while preserving physical consistency. Across canonical cases, the hybrid ML-EPM framework yields substantially tighter, better-calibrated uncertainty estimates than baseline EPM alone.

[94] [A Deep Learning Approach For Epistemic Uncertainty Quantification Of Turbulent Flow Simulations](#)

Minghan Chu and Weicheng Qian. ArXiv, 2024. 0 citations.

0% Topic Match

Abstract: Simulations of complex turbulent flow are part and parcel of the engineering design process. Eddy viscosity based turbulence models represent the workhorse for these simulations. The underlying simplifications in eddy viscosity models make them computationally inexpensive but also introduce structural uncertainties in their predictions. Currently the Eigenspace Perturbation Method is the only approach to predict these uncertainties. Due to its purely physics based nature this method often leads to unrealistically large uncertainty bounds that lead to exceedingly conservative designs. We use a Deep Learning based approach to address this issue. We control the perturbations using trained deep learning models that...

[95] [Physics-constrained polynomial chaos expansion for scientific machine learning and uncertainty quantification](#)

H. Sharma, ..., and Michael D. Shields. ArXiv, 2024. 29 citations.

0% Topic Match

Abstract: We present a novel physics-constrained polynomial chaos expansion as a surrogate modeling method capable of performing both scientific machine learning (SciML) and uncertainty quantification (UQ) tasks. The proposed method possesses a unique capability: it seamlessly integrates SciML into UQ and vice versa, which allows it to quantify the uncertainties in SciML tasks effectively and leverage SciML for improved uncertainty assessment during UQ-related tasks. The proposed surrogate model can effectively incorporate a variety of physical constraints, such as governing partial differential equations (PDEs) with associated initial and boundary conditions constraints, inequality-type constraints (e.g., monotonicity, convexity, non-negativity, among others), and additional a...

[96] [Identifying metamorphic relations: A data mutation directed approach](#)

Chang-ai Sun, ..., and W. Chan. Software: Practice and Experience, 2023. 4 citations.

0% Topic Match

Abstract: Metamorphic testing (MT) is an effective technique to alleviate the test oracle problem. The principle of MT is to detect failures by checking whether some necessary properties, commonly known as metamorphic relations (MRs), of software under test (SUT) hold among multiple executions of source and follow up test cases. Since both the generation of follow up test cases and test result verification depend on MRs, the identification of MRs plays a key role in MT, which is an important yet difficult task requiring deep domain knowledge of the SUT. Accordingly, techniques that can direct a tester to identify MRs effectively are desirable....

[97] [Synergies between the constitutive relation error concept and PGD model reduction for simplified V&V procedures](#)

L. Chamoin, ..., and B. Marchand. Advanced Modeling and Simulation in Engineering Sciences, 2016. 21 citations.

0% Topic Match

Abstract: The paper deals with the constitutive relation error (CRE) concept which has been widely used over the last 40 years for verification and validation of computational mechanics models. It more specifically focuses on the beneficial use of model reduction based on proper generalized decomposition (PGD) into this CRE concept. Indeed, it is shown that a PGD formulation can facilitate the construction of so-called admissible fields which is a technical key-point of CRE. Numerical illustrations, addressing both model verification and model updating, are presented to assess the performances of the proposed approach.

[98] [Adaptivity and Uncertainty: Towards Rigorous Verification and Validation of Flow Simulations](#)

D. Pelletier. Unknown journal, 2010. 0 citations.

0% Topic Match

No summary or abstract available

[99] [Metamorphic Runtime Checking of Applications without Test Oracles](#)

Christian Murphy, ..., and Fang-Hsiang Su. Unknown journal, 2013. 8 citations.

0% Topic Match

No summary or abstract available

[100] [Metamorphic Testing for Smart Contract Validation: A Case Study of Ethereum-Based Crowdfunding Contracts](#)

Irving Jared Villanueva, ..., and Faqeer ur Rehman. 2025 IEEE International Conference on Software Analysis, Evolution and Reengineering - Companion (SANER-C), 2025. 0 citations.

0% Topic Match

Abstract: Blockchain smart contracts play a crucial role in automating and securing agreements in diverse domains such as finance, healthcare, and supply chains. Despite their critical applications, testing these contracts often receives less attention than their development, leaving significant risks due to the immutability of smart contracts post-deployment. A key challenge in the testing of smart contracts is the oracle problem, where the exact expected outcomes are not well defined, complicating systematic testing efforts. Metamorphic Testing (MT) addresses the oracle problem by using Metamorphic Relations (MRs) to validate smart contracts. MRs define how output should change relative to specific input modifications, determining...

[101] [Continuation Multilevel Monte Carlo Evolutionary Algorithm for Robust Aerodynamic Shape Design](#)

M. Pisaroni, ..., and P. Leyland. Journal of Aircraft, 2019. 9 citations.

0% Topic Match

Abstract: The majority of problems in aircraft production and operation require decisions made in the presence of uncertainty. For this reason, aerodynamic designs obtained with traditional deterministic optimization techniques seeking only optimality in a specific set of conditions may have very poor off-design performances or may even be unreliable. In this work, a novel approach for robust and reliability-based design optimization of aerodynamic shapes based on the combination of single- and multi-objective evolutionary algorithms and a continuation multilevel Monte Carlo methodology is presented, to compute objective functions and constraints that involve statistical moments or statistical quantities, such as quantiles, also called...

[102] [Synergies between the constitutive relation error concept and PGD model reduction for simplified V&V procedures](#)

L. Chamoin, ..., and B. Marchand. Advanced Modeling and Simulation in Engineering Sciences, 2016. 0 citations.

0% Topic Match

Abstract: The paper deals with the constitutive relation error (CRE) concept which has been widely used over the last 40 years for verification and validation of computational mechanics models. It more specifically focuses on the beneficial use of model reduction based on proper generalized decomposition (PGD) into this CRE concept. Indeed, it is shown that a PGD formulation can facilitate the construction of so-called admissible fields which is a technical key-point of CRE. Numerical illustrations, addressing both model verification and model updating, are presented to assess the performances of the proposed approach.

[103] [Use of Regression to Extend ASME V&V20 Validation Metrics from Validation Points to an Application Point](#)

L. Peltier, ..., and Luís Eça. ASME 2025 Verification, Validation, and Uncertainty Quantification Symposium, 2025. 0 citations.

0% Topic Match

Abstract: Simulations based on the numerical solution of mathematical/computational models using the laws of physics have become common practice in Engineering. Establishing confidence in the quantities of interest obtained from such simulations requires the assessment of the numerical and modeling accuracy, which is the focus of Verification (numerical accuracy) and Validation (modeling accuracy). The ASME V&V20 standard presents a validation metric that estimates an interval that should contain the modeling error of a scalar quantity of interest at a validation setpoint. The experiment that provides the measurement of the physical reality defines the validation setpoint. The interval that bounds the...

[104] [Unit and regression tests of scientific software: A study on SWMM](#)

Zedong Peng, ..., and Nan Niu. Journal of computational science, 2021. 21 citations.

0% Topic Match

Abstract: Testing helps assure software quality by executing a program and uncovering bugs. Scientific software developers often find it challenging to carry out systematic and automated testing due to reasons like inherent model uncertainties and complex floating-point computations. Extending the recent work on analyzing the unit tests written by the developers of the Storm Water Management Model (SWMM) [32], we report in this paper the investigation of both unit and regression tests of SWMM. The results show that the 2953 unit tests of SWMM have a 39.7% statement-level code coverage and a 82.4% user manual coverage. Meanwhile, an examination of 58...

[105] [Syntax based metamorphic relation prediction via the bagging framework](#)

Yuyu Yin, ..., and Zhijin Pan. Expert Systems, 2021. 4 citations.

0% Topic Match

Abstract: Software testing is an indispensable part of the software engineering industry, which guarantees product reliability and safety. Traditional testing approaches face the testing Oracle problem, they are difficult to construct the expected outputs with the increasing of program complexity. As a result, metamorphic testing, which tests the program by examining the relationship between the execution results, is proposed. However, existing manual metamorphic relation construction requires huge effects of domain experts, and automatic methods are unstable and inefficient due to the insufficient software feature mining. Hence, we proposed a multi dimensional program structure based metamorphic relation prediction approach, which is composed of feature...

[106] [Metamorphic Testing for Investigation of Context Recognition from Smart Home Voice Commands](#)

Fahima Hasan Athina, ..., and Razib Iqbal. 2024 7th Conference on Cloud and Internet of Things (CIoT), 2024. 1 citations.

0% Topic Match

Abstract: The integration of voice control into the Internet of Things (IoT) ecosystem marks a significant advancement, enabling automation through smart assistants such as Amazon Alexa, Google Assistant, and Apple Siri. These assistants rely on neural networks and machine learning models to interpret and execute commands, necessitating high robustness and reliability to avoid errors. However, their unclear internal decision-making mechanism poses challenges in distinguishing between the correct and incorrect behavior of the model, known as the 'oracle problem', complicating testing and verification. Metamorphic testing has emerged as a solution, effectively evaluating and refining machine learning models. In this paper, we present...

[107] [A Framework for Testing Code in Computational Applications](#)

D. Kelly, ..., and Rebecca Sanders. Unknown journal, 2012. 3 citations.

0% Topic Match

No summary or abstract available

[108] [Automated inference of likely metamorphic relations for model transformations](#)

J. Troya, ..., and Antonio Ruiz-Cortés. J. Syst. Softw., 2018. 47 citations.

0% Topic Match

Abstract: Approach to automatically infer likely metamorphic relations in model transformations. A catalogue of 24 domain-independent likely metamorphic relations has been proposed. Five domain-independent patterns in the transformations traces have been identified. Specific metamorphic relations are automatically inferred in each transformation. Precision of 96.4% with 7 case studies and mutation score of 93.3% in one of them. Model transformations play a cornerstone role in Model-Driven Engineering (MDE) as they provide the essential mechanisms for manipulating and transforming models. Checking whether the output of a model transformation is correct is a manual and error-prone task, referred to as the oracle problem. Metamorphic testing alleviates the oracle...

[109] [Assessing the Potential Robustness of Conceptual Rainfall Runoff Models Under a Changing Climate](#)

D. Guo, ..., and L. Marshall. Water Resources Research, 2018. 32 citations.

0% Topic Match

Abstract: Conceptual rainfall runoff (CRR) models are commonly used to assess the potential impact of climate change on water resources systems. However, they are often characterized by poorer performance when used to simulate a different climate compared to that of the calibration period. This is generally referred to as low model robustness, and these issues have been thoroughly explored using historical data. However, the implications of robustness are unknown for a changing climate where models may have to operate under conditions that lie beyond existing observations. This study extends these ideas to evaluate the "potential robustness" of different CRR models in the...

[110] [A Replication Study on Predicting Metamorphic Relations at Unit Testing Level](#)

Alejandra Duque-Torres, ..., and Claus Klammer. 2022 IEEE International Conference on Software Analysis, Evolution and Reengineering (SANER), 2022. 7 citations.

0% Topic Match

Abstract: Metamorphic Testing (MT) addresses the test oracle problem by examining the relations between inputs and outputs of test executions. Such relations are known as Metamorphic Relations (MRs). In current practice, identifying and selecting suitable MRs is usually a challenging manual task, requiring a thorough grasp of the SUT and its application domain. Thus, Kanewala et al. proposed the Predicting Metamorphic Relations (PMR) approach to automatically suggest MRs from a list of six pre-defined MRs for testing newly developed methods. PMR is based on a classification model trained on features extracted from the control-flow graph (CFG) of 100 Java methods. In...

[111] [A non-parametric regularization framework for surrogate learning based on deep encoding](#)

Laura Balasso and A. Alaia. Physics of Fluids, 2026. 0 citations.

0% Topic Match

Abstract: Many engineering applications rely on numerical methods to simulate complex physical systems, which often entail prohibitive computational costs. Additionally, problems involving parametrized domains require the generation of a computational mesh for each configuration, which is an expensive and error-prone process. Surrogates based on proper orthogonal decomposition can alleviate this burden providing rapid predictions of partial differential equations (PDEs) solutions, but they rely on explicit geometric parametrizations and shared reference meshes, requirements rarely satisfied in practice. In this paper, we present a machine-learning-based workflow for predicting PDEs solutions and derived quantities in the context of geometric parametrization, without explicit knowledge of...

[112] [Automating Data-Driven Modeling and Analysis for Engineering Applications using Large Language Model Agents](#)

Yang Liu, ..., and Doyeong Lim. ArXiv, 2025. 4 citations.

0% Topic Match

Abstract: Modern engineering increasingly relies on vast datasets generated by experiments and simulations, driving a growing demand for efficient, reliable, and broadly applicable modeling strategies. There is also heightened interest in developing data-driven approaches, particularly neural network models, for effective prediction and analysis of scientific datasets. Traditional data-driven methods frequently involve extensive manual intervention, limiting their ability to scale effectively and generalize to diverse applications. In this study, we propose an innovative pipeline utilizing Large Language Model (LLM) agents to automate data-driven modeling and analysis, with a particular emphasis on regression tasks. We evaluate two LLM-agent frameworks: a multi-agent system featuring...

[113] [Shock-Aware Constrained Optimization of the RAE2822 Transonic Airfoil via a Two-Channel vSDF Surrogate with Closed-Loop CFD Verification](#)

Yuxin Huo, ..., and Xiaoping Ma. Aerospace, 2026. 0 citations.

0% Topic Match

Abstract: Shock-aware aerodynamic shape optimization of transonic airfoils requires surrogate models that capture both integral aerodynamic trends and shock-relevant pressure distribution features. This study addresses drag-oriented optimization of the RAE2822 transonic airfoil under a lift-targeted condition with prediction relative thickness feasibility, rather than strict target pressure inverse design. Each airfoil is parameterized by a 16-dimensional CST vector and mapped to a two-channel vertical signed distance field representation of the upper- and lower-surface Cp curves, from which shock descriptors, including the shock location indicator xs and the pressure jump magnitude Cp, are extracted in a deterministic, implementation-consistent manner. To quantify the reliability...

[114] [Unit Tests of Scientific Software: A Study on SWMM](#)

Zedong Peng, ..., and Nan Niu. Computational Science – ICCS 2020, 2020. 12 citations.

0% Topic Match

Abstract: Testing helps assure software quality by executing program and uncovering bugs. Scientific software developers often find it challenging to carry out systematic and automated testing due to reasons like inherent model uncertainties and complex floating point computations. We report in this paper a manual analysis of the unit tests written by the developers of the Storm Water Management Model (SWMM). The results show that the 1,458 SWMM tests have a 54.0% code coverage and a 82.4% user manual coverage. We also observe a "getter-setter-getter" testing pattern from the SWMM unit tests. Based on these results, we offer insights to improve...

[115] [Capture the Flags: Family-Based Evaluation of Agentic LLMs via Semantics-Preserving Transformations](#)

Shahin Honarvar, ..., and Alastair Donaldson. ArXiv, 2026. 2 citations.

0% Topic Match

Abstract: Agentic large language models (LLMs) are increasingly evaluated on cybersecurity tasks using capture-the-flag (CTF) benchmarks, yet existing pointwise benchmarks offer limited insight into agent robustness and generalisation across alternative versions of the source code. We introduce CTF challenge families, whereby a single CTF is used to generate a family of semantically-equivalent challenges via semantics-preserving program transformations, enabling controlled evaluation of robustness while keeping the underlying exploit strategy fixed. We present Evolve-CTF, a tool that generates CTF families from Python challenges using a range of transformations. Using Evolve-CTF to derive families from Cybench and Intercode challenges, we evaluate 13 agentic LLM...

[116] [On the Adversarial Robustness of Hydrological Models](#)

Yang Yang, ..., and Ting Fong May Chui. ArXiv, 2026. 1 citations.

0% Topic Match

Abstract: The evaluation of hydrological models is essential for both model selection and reliability assessment. However, simply comparing predictions to observations is insufficient for understanding the global landscape of model behavior. This is especially true for many deep learning models, whose structures are complex. Further, in risk-averse operational settings, water managers require models that are trustworthy and provably safe, as non-robustness can put our critical infrastructure at risk. Motivated by the need to select reliable models for operational deployment, we introduce and explore adversarial robustness analysis in hydrological modeling, evaluating whether small, targeted perturbations to meteorological forcings induce substantial changes in...

[117] [Calculation coverage testing in scientific applications](#)

Yoshiki Sato, ..., and S. Chiba. Proceedings of the 2015 International Symposium on Software Testing and Analysis, 2015. 0 citations.

0% Topic Match

No summary or abstract available

[118] [Combining split-sample testing and hidden Markov modelling to assess the robustness of hydrological models](#)

Étienne Guilpart, ..., and F. Anctil. Hydrology and Earth System Sciences, 2021. 15 citations.

0% Topic Match

Abstract: Abstract. The impacts of climate and land-use changes make the stationary assumption in hydrology obsolete. Moreover, there is still considerable uncertainty regarding the future evolution of the Earth's climate and the extent of the alteration of flow regimes. Climate change impact assessment in the water sector typically involves a modelling chain in which a hydrological model is needed to generate hydrologic projections from climate forcings. Considering the inherent uncertainty of the future climate, it is crucial to assess the performance of the hydrologic model over a wide range of climates and their corresponding hydrologic conditions. In this paper, numerous, contrasted,...

[119] [The acceptance, use, and perceptions of metamorphic testing for a sample of open-source software developers](#)

Brittany R. Hoard. Cogent Engineering, 2025. 2 citations.

0% Topic Match

Abstract: Abstract This article examines the use, acceptance, and perceptions of metamorphic testing (MT) for a sample of open-source software developers. The study findings, including descriptive data and the results of correlational analyses, are used to inform recommendations for practice and for future research. The guiding framework of this study is the unified theory of acceptance and use of technology (UTAUT). An online survey instrument was used to collect data from a sample of contributors to a variety of projects in a GitHub collection. The survey included quantitative questions designed to measure the following UTAUT constructs and variables as they pertain...

[120] [MetaLogic: Robustness Evaluation of Text-to-Image Models via Logically Equivalent Prompts](#)

Yifan Shen, ..., and Yulei Sui. Unknown journal, 2025. 1 citations.

0% Topic Match

Abstract: Recent advances in text-to-image (T2I) models, especially diffusion-based architectures, have significantly improved the visual quality of generated images. However, these models continue to struggle with a critical limitation: maintaining semantic consistency when input prompts undergo minor linguistic variations. Despite being logically equivalent, such prompt pairs often yield misaligned or semantically inconsistent images, exposing a lack of robustness in reasoning and generalisation. To address this, we propose MetaLogic, a novel evaluation framework that detects T2I misalignment without relying on ground truth images. MetaLogic leverages metamorphic testing, generating image pairs from prompts that differ grammatically but are semantically identical. By directly comparing...

[121] [Physics-Preserving Evolutionary Operators for Symbolic Regression](#)

Yunpeng Gong, ..., and Min Jiang. 2025 International Conference on Machine Intelligence and Nature-Inspired Computing (MIND), 2025. 0 citations.

0% Topic Match

Abstract: Symbolic regression via Genetic Programming (GP) has gained popularity for its ability to produce interpretable, closed-form expressions that fit empirical data. However, conventional GP methods often generate expressions that violate fundamental physical principles or produce numerically unstable results, which significantly limits their applicability in scientific modeling. In this work, we propose a novel physics-informed symbolic regression framework that integrates physical laws into the evolutionary search process. Our method incorporates three key innovations: (1) a physics-augmented fitness function that penalizes violations of known physical constraints (e.g., conservation laws, dimensional balance), (2) physics-preserving genetic operators that ensure dimensional consistency and mathematical validity...

[122] [Domain Adaptation for Fault Detection in Civil Nuclear Plants](#)

H. Wood, ..., and Will Jacobs. PHM Society European Conference, 2024. 0 citations.

0% Topic Match

Abstract: Recent domain adaptation approaches have been shown to generalise well between distant data domains achieving high performance in machine fault detection through time series classification. An interesting aspect of this transfer-learning inspired approach, is that the algorithm need not be exposed to fault data from the target domain during training. This promotes the application of these methods to environments in which fault data is unfeasible to obtain, such as the detection of loss-of-coolant accidents (LOCA) in nuclear power plants (NPPs). A LOCA is a failure mode of a nuclear reactor in which coolant is lost due to a physical break...

[123] [Hierarchical Surrogate-Assisted Evolutionary Multi-Scenario Airfoil Shape Optimization](#)

Handing Wang, ..., and Yaochu Jin. 2018 IEEE Congress on Evolutionary Computation (CEC), 2018. 23 citations.

0% Topic Match

Abstract: For multi-scenario airfoil shape optimization problems, an evaluation of a single airfoil is based on its full-scenario drag landscape. To obtain the full-scenario drag landscape, a large number of computational fluid dynamic simulations for different operating conditions must be conducted. Since a single computational fluid dynamic simulation is often time-consuming, evaluations for multi-scenario airfoil shape optimization will be computationally highly intensive. Although surrogate-assisted evolutionary algorithms have been widely applied to expensive optimization problems, existing surrogate-assisted evolutionary algorithms cannot be directly applied to multi-scenario airfoil shape optimization due to the lack of training data. Instead of using surrogate models to directly...

[124] [Faulty Requirements Made Valuable: On the Role of Data Quality in Deep Learning](#)

Harshitha Challa, ..., and Reese Johnson. 2020 IEEE Seventh International Workshop on Artificial Intelligence for Requirements Engineering (AIRE), 2020. 34 citations.

0% Topic Match

Abstract: Large collections of data help evolve deep learning into the state-of-the-art in solving many artificial intelligence problems. However, the requirements engineering (RE) community has yet to adapt to such sweeping changes caused exclusively by data. One reason is that the traditional requirements quality like unambiguity becomes less applicable to data, and so do requirements fault detection techniques like inspections. In this paper, we view deep learning as a class of machines whose effects must be evaluated with direct consideration of inherent data quality attributes: accuracy, consistency, currentness, etc. We substantiate this view by altering stationarity of the multivariate time-series data...

[125] [Machine Learning-Assisted Airfoil Optimization and Perspectives for Morphing Wing Applications](#)

Walid Abouzoul, ..., and Souad Tayane. E3S Web of Conferences, 2025. 1 citations.

0% Topic Match

Abstract: The design of high-performance airfoils is central to aerodynamic efficiency in aircraft and aerospace systems. Conventional optimization methods often depend on iterative computational fluid dynamics (CFD) simulations or wind-tunnel experiments, both of which are accurate but costly in time and resources. This paper introduces a machine learning (ML)-assisted framework for airfoil optimization that accelerates design while maintaining predictive reliability. A dataset of NACA airfoil profiles is used to train surrogate models capable of estimating aerodynamic coefficients. Bayesian optimization then explores the design space to identify profiles with enhanced lift-to-drag ratios. Preliminary results demonstrate that the framework generates improved geometries compared...

[126] [I/O Associations in Scienti c Software: A Study of SWMM](#)

†. ZedongPeng, ..., and ‡. OmarI.Abdul-Aziz. Journal Not Provided, Unknown year. 1 citations.

0% Topic Match

No summary or abstract available

[127] [Robust State Evaluation and Adaptive Fault Warning for Heat Exchangers: An Uncertainty-Aware Reinforcement Learning Approach](#)

Yuzhen Dang, ..., and Gong Lu. 2025 2nd International Conference on Energy Technology and Electrical Power (ETEP), 2025. 0 citations.

0% Topic Match

Abstract: Chemical heat exchangers often face complex dynamic environments characterized by strong coupling and nonlinear disturbances. Traditional threshold-based methods often struggle with these variable operating conditions, leading to high false alarm rates and delayed responses. To address this, this paper proposes a robust Reinforcement Learning-based State Evaluation and Adaptive Warning Strategy (RL-SEAWS). Unlike static classifiers, the proposed agent dynamically optimizes its warning policy by interacting with a tailored reward mechanism that balances sensitivity and reliability. The model improves upon the Deep Q-Network (DQN) by integrating a temporal dual-attention encoder to capture local-global dependencies and an uncertainty-aware Q-value regularization term to penalize...

[128] [Bridging the Gap on AI-Assisted Scientific Software Development Through Transparency and Traceability](#)

Chaitanya Bhawe, ..., and D. Schwen. ArXiv, 2026. 0 citations.

0% Topic Match

Abstract: The widespread adoption of AI-assisted development in scientific software is not a future concern -- it is a present reality. Researchers are already using large language models to write code, generate test cases, and draft documentation, yet this practice remains largely unacknowledged and unguided in formal workflows and published work. This ad hoc, ungoverned use of AI represents a systemic risk to scientific software quality, particularly in safety-relevant modeling and simulation tools subject to strict Software Quality Assurance (SQA), or even Nuclear Quality Assurance Level 1 (NQA-1) standards, for which traceability, independent verification, and documented procedures are paramount. The question...

[129] [Applicability of Software Validation Techniques to Scientific Programs](#)

W. Howden. ACM Transactions on Programming Languages and Systems (TOPLAS), 1980. 40 citations.

0% Topic Match

No summary or abstract available

[130] [Automated continuous verification and validation for numerical simulation](#)

P. Farrell, ..., and C. R. Wilson. Geoscientific Model Development Discussions, 2010. 3 citations.

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Abstract: Introduction Conclusions References

[131] [I/O Associations in Scientific Software: A Study of SWMM](#)

Zedong Peng, ..., and O. Abdul Aziz Unknown journal, 2021. 5 citations.

0% Topic Match

No summary or abstract available

[132] [Unit and Performance Testing of Scientific Software Using MATLAB](#)

B. Koteska, ..., and A. Mishev. Unknown journal, 2019. 0 citations.

0% Topic Match

No summary or abstract available

[133] [End-to-End Deep-Learning-Based Surrogate Modeling for Supersonic Airfoil Shape Optimization](#)

Diogo Pereira, ..., and Fernando Lau. Aerospace, 2025. 7 citations.

0% Topic Match

Abstract: Aerodynamic shape design optimization faces challenges due to the computational demands and the vast design space, limiting its practicality and scalability. While progress has been made in subsonic and transonic regimes, the real-time optimization for supersonic conditions remains unexplored. To bridge this gap, this work exploits knowledge learned from subsonic and transonic real-world data and introduces a rapid optimization framework tailored for the supersonic regime. A novel end-to-end multitask Convolutional Neural Network is proposed to predict the aerodynamic coefficients of an airfoil shape, extracting global and local features directly from the geometry. The surrogate model is thoroughly examined and validated,...

[134] [Releasing Scientific Software in GitHub: A Case Study on SWMM2PEST](#)

Xuanyi Lin, ..., and Nan Niu. 2019 IEEE/ACM 14th International Workshop on Software Engineering for Science (SE4Science), 2019. 9 citations.

0% Topic Match

Abstract: Release engineering involves code development, integration, testing, and software delivery. It has been widely applied to deliver high-quality software to users. While release engineering is a widespread practice in the software industry, there have been very few studies on the release engineering pipeline of scientific software. To shorten this gap, we present a case study in this paper to show a GitHub-driven release workflow on SWMM2PEST, a software system automating parameter calibration for the U.S. EPA's Storm Water Management Model (SWMM). Moreover, we analyze software version updates and requirements changes to develop strategies for improving the ongoing releases. The feasibility...

[135] [Iterative Metamorphic Testing](#)

Peng Wu. 29th Annual International Computer Software and Applications Conference (COMPSAC'05), 2005. 34 citations.

0% Topic Match

No summary or abstract available

[136] [AgentModernize: Preserving Business Logic in Legacy Modernization with Multi-Agent LLMs and Behavioral Specification Graphs](#)

Sheikh Ahmed and Marnim Galib. ArXiv, 2026. 0 citations.

0% Topic Match

Abstract: Legacy modernization breaks business logic. Most tools and LLM-based approaches treat modernization as syntax translation, losing implicit rules, edge-case handling, and cross-module constraints. We present AgentModernize, a multi-agent framework that treats modernization as a behavioral preservation problem. Four specialized agents handle extraction, specification, code generation, and validation. The key intermediate artifact -- a Behavioral Specification Graph (BSG) -- forces extracted business logic to be explicit and inspectable before any code is generated. We evaluated on LegacyModernize-8, eight scenarios spanning telecom and banking, using three models (GPT-4o-mini, GPT-4o, GPT-5.3-codex) under a fair protocol: same gold-standard tests, 3 trials, temperature 0.0. Full...

[137] [Symmetry-Informed Governing Equation Discovery](#)

Jianwei Yang, ..., and Rose Yu. ArXiv, 2024. 10 citations.

0% Topic Match

Abstract: Despite the advancements in learning governing differential equations from observations of dynamical systems, data-driven methods are often unaware of fundamental physical laws, such as frame invariance. As a result, these algorithms may search an unnecessarily large space and discover less accurate or overly complex equations. In this paper, we propose to leverage symmetry in automated equation discovery to compress the equation search space and improve the accuracy and simplicity of the learned equations. Specifically, we derive equivariance constraints from the time-independent symmetries of ODEs. Depending on the types of symmetries, we develop a pipeline for incorporating symmetry constraints into various...

[138] [Metamorphic Testing: Testing the Untestable](#)

Sergio Segura, ..., and T. Chen. IEEE Software, 2020. 109 citations.

0% Topic Match

Abstract: What if we could know that a program is buggy, even if we could not tell whether or not its observed output is correct? Metamorphic testing provides this ability. This article explains the basics of the technique.

[139] [Agentic AI Modernization: Transforming Institutional Infrastructure Through Orchestrated Multi-Agent LLM Framework](#)

Mahesh Kumar Damarched. Journal of Computer Science and Technology Studies, 2026. 2 citations.

0% Topic Match

Abstract: While managing constrained funds and strict regulatory requirements, the higher education institutions are under unprecedented pressure to modernize outdated information systems, such as mainframe-based Student Information Systems (SIS), custom registration platforms, legacy Learning Management Systems (LMS) and Enterprise Resource Planning (ERP) deployments. The complexity of institutional governance is being overlooked by the conventional single-agent based approaches to legacy modernization, which is delaying the digital transformation and creating security vulnerabilities. In order to achieve end-to-end code analysis, intelligent planning, safe migration and rigorous validation through specialized agents coordinated by institutional governance patterns, this research presents a novel agentic architecture using multi-agent...

[140] [On the effectiveness of metamorphic testing for numerical programs](#)

Jianqiang Feng and @Unknown journal, 2003. 3 citations.

0% Topic Match

No summary or abstract available

[141] [Testing and Validating Machine Learning Classifiers by Metamorphic Testing](#)

Xiaoyuan Xie, ..., and T. Chen. The Journal of systems and software, 2011. 398 citations.

Not measured Topic Match

Abstract: Machine Learning algorithms have provided core functionality to many application domains - such as bioinformatics, computational linguistics, etc. However, it is difficult to detect faults in such applications because often there is no "test oracle" to verify the correctness of the computed outputs. To help address the software quality, in this paper we present a technique for testing the implementations of machine learning classification algorithms which support such applications. Our approach is based on the technique "metamorphic testing", which has been shown to be effective to alleviate the oracle problem. Also presented include a case study on a real-world machine...

[142] [On Testing Non-Testable Programs](#)

E. Weyuker. Comput. J., 1982. 643 citations.

Not measured Topic Match

No summary or abstract available

[143] [Metamorphic relation prioritization for effective regression testing](#)

Madhusudan Srinivasan and Upulee Kanewala. Software Testing, 2021. 18 citations.

Not measured Topic Match

Abstract: Metamorphic testing (MT) is widely used for testing programs that face the oracle problem. It uses a set of metamorphic relations (MRs), which are relations among multiple inputs and their corresponding outputs to determine whether the program under test is faulty. Typically, MRs vary in their ability to detect faults in the program under test, and some MRs tend to detect the same set of faults. In this paper, we propose approaches to prioritize MRs to improve the efficiency and effectiveness of MT for regression testing. We present two MR prioritization approaches: (i) fault based and (ii) coverage based. To evaluate these...

[144] [Metamorphic Testing for Quality Assurance of Protein Function Prediction Tools](#)

Morteza Pourreza Shahri, ..., and Upulee Kanewala. 2019 IEEE International Conference On Artificial Intelligence Testing (AITest), 2019. 16 citations.

Not measured Topic Match

Abstract: Proteins are the workhorses of life and gaining insight on their functions is of paramount importance for applications such as drug design. However, the experimental validation of functions of proteins is highly-resource consuming. Therefore, recently, automated protein function prediction (AFP) using machine learning has gained significant interest. Many of these AFP tools are based on supervised learning models trained using existing gold-standard functional annotations, which are known to be incomplete. The main challenge associated with conducting systematic testing on AFP software is the lack of a test oracle, which determines passing or failing of a test case; unfortunately, due to...

[145] [Using machine learning techniques to detect metamorphic relations for programs without test oracles](#)

Upulee Kanewala and J. Bieman. 2013 IEEE 24th International Symposium on Software Reliability Engineering (ISSRE), 2013. 105 citations.

Not measured Topic Match

No summary or abstract available

[146] [Quality Assurance of Bioinformatics Software: A Case Study of Testing a Biomedical Text Processing Tool Using Metamorphic Testing](#)

Madhusudan Srinivasan, ..., and Indika Kahanda. 2018 IEEE/ACM 3rd International Workshop on Metamorphic Testing (MET), 2018. 29 citations.

Not measured Topic Match

Abstract: Bioinformatics software plays a very important role in making critical decisions with in many areas including medicine and healthcare. However, most of the research is directed towards developing tools, and little time and effort is spent on testing the software to assure its quality. In testing, a test oracle is used to determine whether a test is passed or failed during testing, and unfortunately, for much of bioinformatics software, the exact expected outcomes are not well defined. Thus, the main challenge associated with conducting systematic testing on bioinformatics software is the oracle problem. Metamorphic testing (MT) is a technique used...

[147] [Search-based inference of polynomial metamorphic relations](#)

Jie M. Zhang, ..., and Hong Mei. Proceedings of the 29th ACM/IEEE International Conference on Automated Software Engineering, 2014. 118 citations.

Not measured Topic Match

No summary or abstract available

[148] [The Oracle Problem in Software Testing: A Survey](#)

Earl T. Barr, ..., and S. Yoo. IEEE Transactions on Software Engineering, 2015. 1079 citations.

Not measured Topic Match

Abstract: Testing involves examining the behaviour of a system in order to discover potential faults. Given an input for a system, the challenge of distinguishing the corresponding desired, correct behaviour from potentially incorrect behavior is called the "test oracle problem". Test oracle automation is important to remove a current bottleneck that inhibits greater overall test automation. Without test oracle automation, the human has to determine whether observed behaviour is correct. The literature on test oracles has introduced techniques for oracle automation, including modelling, specifications, contract-driven development and metamorphic testing. When none of these is completely adequate, the final source of test...

[149] [Identifying implementation bugs in machine learning based image classifiers using metamorphic testing](#)

Anurag Dwarakanath, ..., and Sanjay Podder. Proceedings of the 27th ACM SIGSOFT International Symposium on Software Testing and Analysis, 2018. 196 citations.

Not measured Topic Match

Abstract: We have recently witnessed tremendous success of Machine Learning (ML) in practical applications. Computer vision, speech recognition and language translation have all seen a near human level performance. We expect, in the near future, most business applications will have some form of ML. However, testing such applications is extremely challenging and would be very expensive if we follow today's methodologies. In this work, we present an articulation of the challenges in testing ML based applications. We then present our solution approach, based on the concept of Metamorphic Testing, which aims to identify implementation bugs in ML based image classifiers. We...

[150] [Validation of Machine Learning Classifiers Using Metamorphic Testing and Feature Selection Techniques](#)

Sadam Al-Azani and Jameleddine Hassine. Unknown journal, 2017. 10 citations.

Not measured Topic Match

No summary or abstract available

[151] [Validating a Deep Learning Framework by Metamorphic Testing](#)

Junhua Ding, ..., and Xin-Hua Hu. 2017 IEEE/ACM 2nd International Workshop on Metamorphic Testing (MET), 2017. 75 citations.

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[152] [The evaluation of program-based software test data adequacy criteria](#)

E. Weyuker. Communications of the ACM, 1988. 169 citations.

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[153] [Simulation validation using metamorphic testing \(WIP\)](#)

M. Raunak and Megan M. Olsen. Unknown journal, 2015. 8 citations.

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[154] [Metamorphic Model-Based Testing of Autonomous Systems](#)

M. Lindvall, ..., and Christoph Schulze. 2017 IEEE/ACM 2nd International Workshop on Metamorphic Testing (MET), 2017. 64 citations.

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[155] [Early Results from Metamorphic Testing of Epidemiological Models](#)

L. Pullum and Ozgur Ozmen. 2012 ASE/IEEE International Conference on BioMedical Computing (BioMedCom), 2012. 34 citations.

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[156] [On effective testing of health care simulation software](#)

Christian Murphy, ..., and L. Osterweil. Unknown journal, 2011. 36 citations.

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[157] [Conformance Testing of Network Simulators Based on Metamorphic Testing Technique](#)

T. Chen, ..., and Shengqiong Wang. Unknown journal, 2009. 29 citations.

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[158] [Metamorphic Testing: A Simple Method for Alleviating the Test Oracle Problem](#)

T. Chen. 2015 IEEE/ACM 10th International Workshop on Automation of Software Test, 2015. 29 citations.

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[159] [A New Method for Constructing Metamorphic Relations](#)

Huai Liu, ..., and T. Chen. 2012 12th International Conference on Quality Software, 2012. 87 citations.

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[160] [Application of metamorphic testing monitored by test adequacy in a Monte Carlo simulation program](#)

Junhua Ding and Xin-Hua Hu. Software Quality Journal, 2016. 9 citations.

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[161] [A mapping study on testing non-testable systems](#)

Krishna Patel and R. Hierons. Software Quality Journal, 2017. 35 citations.

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Abstract: The terms "Oracle Problem" and "Non-testable system" interchangeably refer to programs in which the application of test oracles is infeasible. Test oracles are an integral part of conventional testing techniques; thus, such techniques are inoperable in these programs. The prevalence of the oracle problem has inspired the research community to develop several automated testing techniques that can detect functional software faults in such programs. These techniques include N-Version testing, Metamorphic Testing, Assertions, Machine Learning Oracles, and Statistical Hypothesis Testing. This paper presents a Mapping Study that covers these techniques. The Mapping Study presents a series of discussions about each technique,...

[162] [Multi-objective optimization of cloud systems](#)

M. Perez, ..., and Alberto Nuñez. Sci. Comput. Program., 2026. 0 citations.

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[163] [Identification Algorithm Framework and Structural model on Input Pattern of Metamorphic Relations](#)

Shiyu Yan, ..., and Jie Liu. 2022 9th International Conference on Dependable Systems and Their Applications (DSA), 2022. 1 citations.

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Abstract: Metamorphic Relation (MR) plays a crucial role in Metamorphic Testing (MT), which can be regarded as the implication relation from the input pattern to the output pattern. Based on MR's properties, the identification of Metamorphic Relations (MRs) can be from its input pattern and output pattern, respectively. However, the finding of the effective input pattern is very first and important. Especially for complicated mathematical function includes more than two input variables, we systematically analyze the structural model of input pattern in MR, propose the concept of the class of basic input pattern, as well as prove that any one input...

[164] [A partial oracle for uniformity statistics](#)

Krishna Patel and R. Hierons. Software Quality Journal, 2019. 3 citations.

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Abstract: This paper investigates the problem of testing implementations of uniformity statistics. In this paper, we used metamorphic testing to address the oracle problem of checking the output of one or more test executions, for uniformity statistics. We defined a partial oracle that uses regression analysis (a regression model-based metamorphic relation). We investigated the effectiveness of our partial oracle. We found that the technique can achieve mutation scores ranging from 77.78 to 100% and tends towards higher mutation scores in this range. These results are promising and suggest that the regression model-based metamorphic relation approach is a viable method of alleviating...

[165] [Refinement type contracts for verification of scientific investigative software](#)

Maxwell Shinn. ArXiv, 2019. 4 citations.

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Abstract: Our scientific knowledge is increasingly built on software output. User code which defines data analysis pipelines and computational models is essential for research in the natural and social sciences, but little is known about how to ensure its correctness. The structure of this code and the development process used to build it limit the utility of traditional testing methodology. Formal methods for software verification have seen great success in ensuring code correctness but generally require more specialized training, development time, and funding than is available in the natural and social sciences. Here, we present a Python library which uses lightweight...

[166] [Challenges for Verifying and Validating Scientific Software in Computational Materials Science](#)

T. Vogel, ..., and L. Grunske. 2019 IEEE/ACM 14th International Workshop on Software Engineering for Science (SE4Science), 2019. 14 citations.

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Abstract: Many fields of science rely on software systems to answer different research questions. For valid results researchers need to trust the results scientific software produces, and consequently quality assurance is of utmost importance. In this paper we are investigating the impact of quality assurance in the domain of computational materials science (CMS). Based on our experience in this domain we formulate challenges for validation and verification of scientific software and their results. Furthermore, we describe directions for future research that can potentially help dealing with these challenges.

[167] [Black-Box Testing for Practitioners: A Case of the New ISTQB Test Analyst Syllabus](#)

Matthias Hamburg and Adam Roman. 2025 IEEE Conference on Software Testing, Verification and Validation (ICST), 2025. 0 citations.

Not measured Topic Match

Abstract: The International Software Testing Qualifications Board (ISTQB) is a volunteer organization aiming to qualify software testing and quality assurance practitioners. It issues a certification scheme with many certification products for this subject matter. These products are based on the current state of practice and consist of a syllabus and a set of sample exam questions. They can also serve as a basis for academic courses in software testing. We recently revised and updated the Advanced Level Test Analyst Syllabus, which focuses on black-box test techniques. In this paper, we share our experience in developing the new syllabus. We describe the...

[168] [Testing high performance numerical simulation programs: experience, lessons learned, and open issues](#)

Xiao He, ..., and Yi Liu. Proceedings of the 29th ACM SIGSOFT International Symposium on Software Testing and Analysis, 2020. 20 citations.

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Abstract: High performance numerical simulation programs are widely used to simulate actual physical processes on high performance computers for the analysis of various physical and engineering problems. They are usually regarded as non-testable due to their high complexity. This paper reports our real experience and lessons learned from testing five simulation programs that will be used to design and analyze nuclear power plants. We applied five testing approaches and found 33 bugs. We found that property-based testing and metamorphic testing are two effective methods. Nevertheless, we suffered from the lack of domain knowledge, the high test costs, the shortage of test...

[169] [Metamorphic Testing for Web System Security](#)

N. B. Chaleshtari, ..., and L. Briand. IEEE Transactions on Software Engineering, 2022. 31 citations.

Not measured Topic Match

Abstract: Security testing aims at verifying that the software meets its security properties. In modern Web systems, however, this often entails the verification of the outputs generated when exercising the system with a very large set of inputs. Full automation is thus required to lower costs and increase the effectiveness of security testing. Unfortunately, to achieve such automation, in addition to strategies for automatically deriving test inputs, we need to address the oracle problem, which refers to the challenge, given an input for a system, of distinguishing correct from incorrect behavior (e.g., the response to be received after a specific HTTP...

[170] [Testing and Quality Validation for AI Software—Perspectives, Issues, and Practices](#)

Chuanqi Tao, ..., and Tiexin Wang. IEEE Access, 2019. 47 citations.

Not measured Topic Match

Abstract: With the fast growth of artificial intelligence and big data computing technologies, more and more software service systems have been developed using diverse machine learning models and technologies to make business and intelligent decisions based on their multimedia input to achieve intelligent features, such as image recognition, recommendation, decision making, prediction, etc. Nevertheless, there are increasing quality problems resulting in erroneous testing costs in enterprises and businesses. Existing work seldom discusses how to perform testing and quality validation for AI software. This paper focuses on quality validation for AI software function features. The paper provides our understanding of AI software...

[171] [Testing bidirectional model transformation using metamorphic testing](#)

Xiao He, ..., and Gang Huang. Inf. Softw. Technol., 2018. 9 citations.

Not measured Topic Match

Abstract: Abstract Context In model-based software development, bidirectional model transformation (BX) provides a fundamental solution to model synchronization that can retain the consistency among models. Similar to conventional programs, a BX program may also contain bugs. Accordingly, a BX program must be tested prior to being used in practice. A challenging problem of testing BX programs is to construct test oracles (e.g., assertions and expected output models), which are usually difficult and/or expensive to manually specify. Objective In we paper, we investigate how to alleviate the oracle problem in BX testing via reducing the costs of developing test oracles. Method We...

[172] [Automatic Detection of Failures and Anomalies Using Metamorphic Relations for Search Engines](#)

K. Sandhya and S. Vasundra. Unknown journal, 2017. 0 citations.

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[173] [Scalability and Validation of Big Data Bioinformatics Software](#)

A. Yang, ..., and J. Ho. Computational and Structural Biotechnology Journal, 2017. 56 citations.

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Abstract: This review examines two important aspects that are central to modern big data bioinformatics analysis – software scalability and validity. We argue that not only are the issues of scalability and validation common to all big data bioinformatics analyses, they can be tackled by conceptually related methodological approaches, namely divide-and-conquer (scalability) and multiple executions (validation). Scalability is defined as the ability for a program to scale based on workload. It has always been an important consideration when developing bioinformatics algorithms and programs. Nonetheless the surge of volume and variety of biological and biomedical data has posed new challenges. We discuss...

[174] [Generation of Source Test Cases in Metamorphic Testing](#)

Amit Verma and Rupinder Kaur. Imperial journal of interdisciplinary research, 2017. 0 citations.

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[175] [Improve the Quality of ARC Systems Based on the Metamorphic Testing](#)

Jihu Zhang, ..., and Yunwei Dong. 2016 International Symposium on System and Software Reliability (ISSSR), 2016. 5 citations.

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[176] [Quality Assurance for Big Data Application - Issues, Challenges, and Needs](#)

Chuanqi Tao and J. Gao. Unknown journal, 2016. 21 citations.

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No summary or abstract available

[177] [Looking for an MR? Try METWiki Today](#)

Xiaoyuan Xie, ..., and T. Chen. 2016 IEEE/ACM 1st International Workshop on Metamorphic Testing (MET), 2016. 10 citations.

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